

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRC14EB003SYNGAS COMP 3rd Stage Intercooler.pvdb

Date of Analysis : Sep 3,2026 11:33am

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 955.000 mm. ID	
Vessel Design Length, Tangent to Tangent	10912.67 mm.
Specified Datum Line Distance	0.00 mm.
Shell Side Design Temperature	200 °C
Channel Side Design Temperature	85 °C
Shell Side Design Pressure	191.000 bars
Channel Side Design Pressure	147.000 bars
Shop Shell Side Test Pressure	249.563 bars
Shop Channel Side Test Pressure	194.035 bars
Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-350 LF2	1	...	K03011	+IT
Flg Bolting	SA-193 B7	...	64 < t <= 100	G41400	
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
CH. COVER-1	147.000	0.00	155.000	3.0000	N/A	No
CH. FLG-1	147.094	0.00	152.000	3.1750	N/A	No
CH. SHELL-1	147.094	0.00	153.000	3.0000	N/A	No
MAIN SHELL	191.094	0.00	194.000	3.0000	N/A	No
CH. SHELL-2	147.094	0.00	153.000	3.0000	N/A	No
CH. FLG-2	147.094	0.00	153.000	3.0000	N/A	No
CH. COVER-2	147.000	0.00	155.000	3.0000	N/A	No

Liquid Level: 955.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Body Flg	268.2	265.0	...	265.0	257.8	...	1.00	1.00
Body Flg	658.2	390.0	...	300.0	293.2	...	1.00	1.00
Cylinder	1858.2	1200.0	60.0	60.0	57.7	...	1.00	1.00
Cylinder	8878.2	6840.0	77.0	77.0	75.6	...	1.00	1.00
Cylinder	10258.2	1200.0	60.0	60.0	57.7	...	1.00	1.00



Body Flg	10647.7	385.0	...	295.0	280.4	...	1.00	1.00
Body Flg	10912.7	265.0	...	265.0	257.8	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
TS OUT NOZ. T2	WRC-297	0.949	Shell	Passed
SH. IN NOZ. S1	WRC-297	0.971	Shell	Passed
SH. OUT NOZ. S	WRC-297	0.971	Shell	Passed
TS. IN NOZ. T1	WRC-297	0.949	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	160.000	mm.
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1400.000	mm.
Wear Pad Width	250.000	mm.
Wear Pad Thickness	18.000	mm.
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1370.000	mm.
Baseplate Length	1090.000	mm.
Baseplate Thickness	32.000	mm.
Baseplate Width	200.000	mm.
Number of Ribs (including outside ribs)	3	
Rib Thickness	18.000	mm.
Web Thickness	18.000	mm.
Height of Center Web	306.000	mm.
Number of Bolts in Baseplate	2	

Baseplate Sketch

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	41207.30	Kgf
Maximum Transverse Saddle Shear Load	2613.55	Kgf
Maximum Longitudinal Saddle Shear Load	5227.10	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	30346.20	Kgf
Maximum Transverse Saddle Shear Load	225.81	Kgf
Maximum Longitudinal Saddle Shear Load	81.72	Kgf

Weights:

Fabricated - Bare w/o removable internals	48089.3	kg.
Shop Test - Fabricated + water (full)	54710.2	kg.
Shipping - Fab. + rem. intls.+ shipping app.	48278.9	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	48278.9	kg.
Empty - Fab. + intls. + details + wghts.	48278.9	kg.
Operating - Empty + operating liquid (no ca)	54896.0	kg.

Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
TS OUT NOZ. T2	149	...	...	OK	26.47	OK	Passed
DRAIN NOZ. D1	153	...	...	OK	6.42	OK	No Calc[*]
SH. IN NOZ. S1	192	...	...	OK	49.46	OK	Passed
SH. OUT NOZ. S2	192	...	...	OK	49.49	OK	Passed
TS. IN NOZ. T1	149	...	...	OK	26.48	OK	Passed
VENT NOZ. V1	153	...	...	OK	6.42	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 192 Nozzle : SH. IN NOZ. S1 [Shellside]
 Minimum MAWP Nozzles : 149 Nozzle : TS. IN NOZ. T1 [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	TS OUT NOZ. T2	1158.175	90.000	824.800
30	DRAIN NOZ. D1	1258.175	-90.000	174.325
40	SH. IN NOZ. S1	2838.175	90.000	1231.200
40	SH. OUT NOZ. S2	8072.175	-90.000	1231.200
50	TS. IN NOZ. T1	9758.175	-90.000	824.800
50	VENT NOZ. V1	9658.175	90.000	174.325

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
DRAIN NOZ. D1	2.000 in	160	LWN	2.375	8.738	...	...	360.95	1500
VENT NOZ. V1	2.000 in	160	LWN	2.375	8.738	...	...	360.95	1500
TS OUT NOZ. T2	16.000 in	Actual	RTJWn	19.188	40.490	...	...	472.78	1500
TS. IN NOZ. T1	16.000 in	Actual	RTJWn	19.188	40.490	...	...	472.78	1500
SH. IN NOZ. S1	24.000 in	Actual	RTJWn	28.724	60.000	...	...	754.50	1500
SH. OUT NOZ. S2	24.000 in	Actual	RTJWn	28.688	59.540	...	...	754.50	1500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
DRAIN NOZ.	SA-350 LF2	60.000	10.000	...	...	...
VENT NOZ. V	SA-350 LF2	8.738	10.000	...	...	...
TS OUT NOZ.	SA-350 LF2	60.000	10.000	...	...	...
TS. IN NOZ.	SA-350 LF2	60.000	10.000	...	...	...
SH. IN NOZ.	SA-350 LF2	77.000	10.000	...	...	...
SH. OUT NOZ	SA-350 LF2	77.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
DRAIN NOZ. D1	1258.175	-90.0	300.00	0.00	CH. SHELL-1
VENT NOZ. V1	9658.175	90.0	300.00	0.00	CH. SHELL-2
TS OUT NOZ. T2	1158.175	90.0	300.00	0.00	CH. SHELL-1
TS. IN NOZ. T1	9758.175	-90.0	300.00	0.00	CH. SHELL-2
SH. IN NOZ. S1	2838.175	90.0	200.00	0.00	MAIN SHELL
SH. OUT NOZ. S2	8072.175	-90.0	200.00	0.00	MAIN SHELL

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	BLIND FLANGE: 265.0mm. THK X 1508.0mm. OD	SA-350 LF2
1	WELD NECK FLANGE: 300.0mm. THK X 1508.0mm. OD	SA-350 LF2
2	CYLINDER: 60.0mm. THK X 955.0mm. ID X 1200.0mm.	SA-516 70
1	CYLINDER: 77.0mm. THK X 955.0mm. ID X 6840.0mm.	SA-516 70
1	WELD NECK FLANGE: 295.0mm. THK X 1508.0mm. OD	SA-350 LF2
16	BAFFLES: 940.0mm DIA X 8.0mm THK	SA-516 70
2	TUBESHEET: 1109.0mm X 180.0mm THK	SA-350 LF2
367	TUBES: 7200.0mm X 25.4mm DIA X 3.4mm THK	SA-179
2	SADDLE: 160.0mm X 120 DEG	SA-516 70
1	INSULATION: 6834mm X 40mm THK	HOT INSULATION
2	CLASS 1500 GR 1.1, 2.0" BLIND FLANGE(S)	SA-350 LF2
2	GASKET: 987.0mm. OD X 961.0mm. ID	...
48	BODY FLANGE BOLTS: 80.0mm. DIA	SA-193 B7
96	NUTS FOR BODY FLANGE BOLTS: 80.0mm. DIA	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
MAIN SHELL [8]		B	26	25		0.986	77.000	1.00	Yes
FIXED SADDLE [24]		B	-27	-28	-29	0.986	18.000	1.00	No
SLIDING SADDL [24]		B	-27	-28	-29	0.986	18.000	1.00	No
SH. IN NOZ. S1 [1]		B	26	25		0.980	77.000	1.00	Yes
Nozzle Flg [5]		!	-46	-60					
SH. OUT NOZ. S [1]		B	26	25		0.981	77.000	1.00	Yes
Nozzle Flg [5]		!	-46	-59					
Nozl. Bolting [22]			-48			1.000	64.000	1.00	No
Tubesheet: SS [13]		B	9	7		0.960	77.000	1.00	No
Warmest MDMT:			26	25					
CH. COVER-1 [11]		!	-46	-46		0.939	66.250	1.00	No
CH. FLG-1 [11]		!	-46	-46		0.926	60.000	1.00	Yes
CH. FLG-2 [11]		!	-46	-46		0.926	60.000	1.00	Yes
CH. COVER-2 [11]		!	-46	-46		0.939	66.250	1.00	No
CH. SHELL-1 [8]		B	21	19		0.975	60.000	1.00	Yes
CH. SHELL-2 [8]		B	21	19		0.975	60.000	1.00	Yes
TS OUT NOZ. T2 [1]		B	21	19		0.959	60.000	1.00	Yes
Nozzle Flg [5]		A	-18	-42					
DRAIN NOZ. D1 [1]		B	-29	-48	-29	0.599	7.645	1.00	No
Nozzle Flg [5]		!	-46	-46					
TS. IN NOZ. T1 [1]		B	21	19		0.960	60.000	1.00	Yes
Nozzle Flg [5]		!	-46	-70					
VENT NOZ. V1 [1]		B	-29	-48	-29	0.598	7.645	1.00	No
Nozzle Flg [5]		!	-46	-46					
Nozl. Bolting [22]			-48			1.000	64.000	1.00	No
Tubesheet: CS [14]		B	4	-9		0.773	60.000	1.00	No
Bolting [21]			-40						
Bolting [21]			-40						
Warmest MDMT:			21	19					
Exchanger Side			Computed MDMT		Required MDMT		Pass/Fail		
Shell			25.0		-28.9		Fail		
Channel/Tube			19.0		-28.9		Fail		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfcmm2mm):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	191	bars
Channel Side Design Pressure	147	bars
Shell Side Design Temperature	200.0	°C
Channel Side Design Temperature	85.0	°C
Radiography, Shell Side	None	
Radiography, Channel Side	None	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS



Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No



Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Flange
Description	CH. COVER-1
Distance "FROM" to "TO"	265 mm.
Flange Outside Diameter	1508 mm.
Element Thickness	265 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	0 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	229.57 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011
Class / Thickness / Grade	1 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Flange
Description	CH. FLG-1
Distance "FROM" to "TO"	390 mm.
Flange Inside Diameter	955 mm.
Element Thickness	300 mm.
Internal Corrosion Allowance	3.175 mm.
Nominal Thickness	0 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	955 mm.
Liquid Density	0.0009984 kg./cm ³



Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	CH. SHELL-1
Distance "FROM" to "TO"	1200 mm.
Inside Diameter	955 mm.
Element Thickness	60 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	60 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	242 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	955 mm.
Liquid Density	0.0009984 kg./cm ³

Element From Node	30
Detail Type	Nozzle
Detail ID	TS OUT NOZ. T2
Dist. from "FROM" Node / Offset dist	500 mm.
Nozzle Diameter	16 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	893.94 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	30
Detail Type	Nozzle
Detail ID	DRAIN NOZ. D1
Dist. from "FROM" Node / Offset dist	600 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	1500
Layout Angle	-90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT



Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	MAIN SHELL
Distance "FROM" to "TO"	6840 mm.
Inside Diameter	955 mm.
Element Thickness	77 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	77 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	FIXED SADDLE
Dist. from "FROM" Node / Offset dist	1370 mm.
Width of Saddle	160 mm.
Height of Saddle at Bottom	1400 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	250 mm.
Thickness of Wear Plate	18 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.6000002
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	40
Detail Type	Saddle
Detail ID	SLIDING SADDLE
Dist. from "FROM" Node / Offset dist	5190 mm.
Width of Saddle	160 mm.
Height of Saddle at Bottom	1400 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	250 mm.
Thickness of Wear Plate	18 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.1
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	955 mm.
Liquid Density	0.0009984 kg./cm³

Element From Node	40
Detail Type	Insulation
Detail ID	HOT
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	6834 mm.
Thickness of Insulation	40 mm.
Density	0.000192 kg./cm³



Element From Node 40
 Detail Type Nozzle
 Detail ID SH. IN NOZ. S1
 Dist. from "FROM" Node / Offset dist 800 mm.
 Nozzle Diameter 24 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 2260.2 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID SH. OUT NOZ. S2
 Dist. from "FROM" Node / Offset dist 6034 mm.
 Nozzle Diameter 24 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle -90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 2260 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element# 5/7
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description CH. SHELL-2
 Distance "FROM" to "TO" 1200 mm.
 Inside Diameter 955 mm.
 Element Thickness 60 mm.
 Internal Corrosion Allowance 3 mm.
 Nominal Thickness 60 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 147 bars
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Liquid
 Detail ID Liquid: 50
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 955 mm.
 Liquid Density 0.0009984 kg./cm³

Element From Node 50
 Detail Type Nozzle
 Detail ID TS. IN NOZ. T1
 Dist. from "FROM" Node / Offset dist 700 mm.
 Nozzle Diameter 16 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle -90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 900.47 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT



Element From Node	50
Detail Type	Nozzle
Detail ID	VENT NOZ. V1
Dist. from "FROM" Node / Offset dist	600 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	1500
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	16.353 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Flange
Description	CH. FLG-2
Distance "FROM" to "TO"	385 mm.
Flange Inside Diameter	955 mm.
Element Thickness	295 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	0 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	229.57 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011
Class / Thickness / Grade	1 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	955 mm.
Liquid Density	0.0009984 kg./cm ³

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Flange
Description	CH. COVER-2
Distance "FROM" to "TO"	265 mm.
Flange Outside Diameter	1508 mm.
Element Thickness	265 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	0 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C



Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

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XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
CH. COVER-1		268.175	...	265	...
CH. FLG-1		658.175	...	390	...
CH. SHELL-1		1858.18	...	1200	...
MAIN SHELL		8878.17	...	6840	...
CH. SHELL-2		10258.2	...	1200	...
CH. FLG-2		10647.7	...	385	...
CH. COVER-2		10912.7	...	265	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values
Description: CH. COVER-1 :
CH. COVER-1

Flange Type		Blind	
Design Pressure	P	147.00	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1508.000	mm.
Flange Thickness	t	265.0000	mm.
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	4272.566	mm.
Diameter of Bolt Circle	C	1360.000	mm.
Nominal Bolt Diameter	a	80.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	990.000	mm.
Flange Face Inside Diameter	Fid	0.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	987.000	mm.
Gasket Inside Diameter	Gi	961.000	mm.
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.

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Gasket Contact Width,	$N = (Go - Gi)/2$	13.000	mm.
Basic Gasket Width,	$bo = N/2$	6.500	mm.
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425	mm.
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	974.151	mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 974.1509^2 \cdot 147.0$$

$$= 1117265.250 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 974.1509 \cdot 3.0 \cdot 147.0$$

$$= 176841.625 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\
 &= \max(1117265 + 176842 + 0, 0) \\
 &= 1294106.875 \text{ Kgf}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.95 \cdot 6.4246 \cdot \pi \cdot 974.151 + 0.0 \cdot 0.0 \cdot 0.0 \\
 &= 138237.203 \text{ Kgf}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((1294107 + 0 + 4 \cdot 0 / 974.15) / 159, 138237 / 159) \\
 &= 798.196 \text{ cm}^2
 \end{aligned}$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	798.196	978.440	
Radial Distance between Bolts and Edge:	74.000	74.000	
Circ. Spacing between the Bolts:	162.600	177.516	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 978.44 \cdot 159.0 / (68.95 \cdot \pi (987.0 + 961.0)) \\
 &= 36.868 \text{ mm. [Note: Exceeds actual gasket width, 13.000]}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a (A_m + A_b) / 2 \\
 &= 159.0 (798.196 + 978.4396) / 2 \\
 &= 1440220.38 \text{ Kgf}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o \\
 &= 1294106.88 \text{ Kgf}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (1360.0 - 974.1509) / 2 \\
 &= 192.9246 \text{ mm.}
 \end{aligned}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$\begin{aligned}
 &= 1.9 \cdot W_{m1} \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2 \\
 &= 1.9 \cdot 1294107 \cdot 192.9246 \cdot 1.0 / (262.0^2 \cdot 974.1509) + 0.3 \cdot 1.0 \cdot 147.0 \cdot 974.1509^2 / 262.0^2 \\
 &= 130.54 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$\begin{aligned}
 &= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G) \\
 &= 1.9 \cdot 1440220 \cdot 192.9246 \cdot 1.0 / (262.0^2 \cdot 974.1509) \\
 &= 77.42 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= W_{m1} / A_b \\
 &= 1294107 / 978.4396 \\
 &= 129.71 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (W_{m2} / A_b) \\
 &= (138237 / 978.4396) \\
 &= 13.86 \text{ N./mm}^2
 \end{aligned}$$

Flange Stress Analysis Results: N./mm²

Actual	Operating Allowed	Actual	Gasket Seating Allowed



Tangential Flange	130.54	138.00	77.42	138.00
Bolting	129.71	159.00	13.86	159.00

Reqd. Blind Flange Thickness at Center 257.820 mm.
 Reqd. Blind Flange Thickness at Gasket 199.246 mm.

Estimated M.A.W.P. (Operating) 155 bars
 Estimated Finished Weight of Flange at given Thk. 3668.3 kg.
 Estimated Unfinished Weight of Forging at given Thk 3668.3 kg.

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Tangential Flange	127.60	138.00	75.68	138.00
Bolting	129.71	159.00	13.86	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 254.838 mm.

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Flange Input Data Values
Description: FLANGE :
CH. FLG-1

Flange Type	Integral Weld Neck		
Design Pressure	P	147.09	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.1750	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	955.000	mm.
Flange Outside Diameter	A	1508.000	mm.
Flange Thickness	t	300.0000	mm.
Thickness of Hub at Small End	go	60.0000	mm.
Thickness of Hub at Large End	g1	70.0000	mm.
Length of Hub	h	90.0000	mm.
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	1360.000	mm.
Nominal Bolt Diameter	a	80.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	990.000	mm.
Flange Face Inside Diameter	Fid	955.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	987.000	mm.
Gasket Inside Diameter	Gi	961.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62.06	N./mm ²
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (147.09 \cdot (955.0/2 + 3.175)) / (138.0 \cdot 1.0 - 0.6 \cdot 147.09) + Ca$$

$$= 57.9149 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 56.825) / (480.675 + 0.6 \cdot 56.825)$$

$$= 152.325 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 60.0) / (477.5 + 0.6 \cdot 60.0)$$

$$= 161.234 \text{ bars}$$

Corroded Flange ID,	Bcor = B+2*ci	961.350 mm.
Corroded Large Hub,	g1Cor = g1-ci	66.825 mm.



Corroded Small Hub, $g0Cor = go-ci$ 56.825 mm.
 Code R Dimension, $R = ((C-Bcor)/2)-g1cor$ 132.500 mm.
 Gasket Contact Width, $N = (Go - Gi)/2$ 12.825 mm.
 Basic Gasket Width, $bo = N/2$ 6.412 mm.
 Effective Gasket Width, $b = Cb*sqrt(bo)$ 6.381 mm.
 Gasket Reaction Diameter, $G = Go - 2 * b$ 974.238 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 974.2377^2 \cdot 147.094$$

$$= 1118175.875 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.3812 \cdot \pi \cdot 974.2377 \cdot 3.75 \cdot 147.09$$

$$= 219718.625 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 961.35^2 \cdot 147.0936$$

$$= 1088788.125 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 1118176 - 1088788$$

$$= 29387.836 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1118176 + 219719 + 0, 0)$$

$$= 1337894.500 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 62.06 \cdot 6.3812 \cdot \pi \cdot 974.238 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 123584.320 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((1337895 + 0 + 4 \cdot 0/974.24)/159, 123584/159)$$

$$= 825.204 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 80.0 + 6 \cdot 300.0/(3.75 + 0.5)$$

$$= 583.529 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1360.0 \cdot \sin(\pi/24)$$

$$= 177.516 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs/(2a + t) }, 1)$$

$$= \max(\sqrt{ 177.516/(2 \cdot 80.0 + 300.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	825.204	978.440	
Radial Distance between Hub and Bolts:	91.500	132.500	
Radial Distance between Bolts and Edge:	74.000	74.000	
Circ. Spacing between the Bolts:	162.600	177.516	583.529

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 978.44 \cdot 159.0 / (62.06 \cdot \pi (987.0 + 961.0))$$

$$= 40.964 \text{ mm. [Note: Exceeds actual gasket width, 13.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 159.0 \cdot 978.4396$$

$$= 1586333.88 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1337895 - 1118176$$

$$= 219718.62 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1360.0 - 974.2377) / 2$$

$$= 192.8811 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1360.0 - 961.35) / 2 + 192.8811)$$

$$= 196.1031 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1360.0 - 961.35 - 66.825) / 2$$

$$= 165.9125 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1088788.	165.9125	1.0000	180647.
Face Pressure, Mt	29388.	196.1031	1.0000	5763.
Gasket Load, Mg	219719.	192.8811	1.0000	42380.
Gasket Seating, Matm	1586334.	192.8811	1.0000	305980.

Total Moment for Operation, Mo 228791. Kg-m.
 Total Moment for Gasket seating, Mg 305980. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 233.728 mm.
 Hub Ratio, h/h0 = HL / H0 0.385
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.176

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.889
 Factor V per 4.16.5 0.448
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1508.0 / 961.35$$

$$= 1.569$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.569^2 \cdot \log_{10}(1.569))) / (1.569^2 - 1)) / (1.569 - 1)$$

$$= 4.487$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.569^2 (1 + 8.55246 \cdot \log_{10}(1.569)) - 1) / ((1.0472 + 1.9448 \cdot 1.569^2) \cdot (1.569 - 1))$$

$$= 1.681$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.569^2(1 + 8.55246 \cdot \log_{10}(1.569)) - 1) / (1.36136(1.569^2 - 1) \cdot (1.569 - 1))$$

$$= 4.931$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.569^2 + 1) / (1.569^2 - 1)$$

$$= 2.369$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 4.931 \cdot 56.825^2 \cdot 233.728 / 0.448$$

$$= 8314152.500 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.889 / 233.728$$

$$= 0.004 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / (T + t^3/d)$$

$$= (300.0 \cdot 0.004 + 1) / (1.681 + 300.0^3 / 8314153)$$

$$= 4.521$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 228791 / 961.35) / (4.5211 \cdot 66.825^2)$$

$$= 115.60 \text{ N/mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 305980 / 961.35) / (4.5211 \cdot 66.825^2)$$

$$= 154.60 \text{ N/mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.5212 \cdot 228791 / 961.35) / (4.5211 \cdot 300.0^2)$$

$$= 14.46 \text{ N/mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.5212 \cdot 305980 / 961.35) / (4.5211 \cdot 300.0^2)$$

$$= 19.34 \text{ N/mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.4871 \cdot 228791 / (300.0^2 \cdot 961.35)) - 2.3693 \cdot 14$$

$$= 82.10 \text{ N/mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.4871 \cdot 305980 / (300.0^2 \cdot 961.35)) - 2.3693 \cdot 19$$

$$= 109.79 \text{ N/mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (116 + \max(14, 82)) / 2$$

$$= 98.85 \text{ N/mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (155 + \max(19, 110)) / 2$$

$$= 132.20 \text{ N/mm}^2$$

Bolt Stress, Operating [BSo]:

= $Wm1 / Ab$
 = 1337895/978.4396
 = 134.10 N./mm²

Bolt Stress, Seating [BSa]:
 = ($Wm2 / Ab$)
 = (123584/978.4396)
 = 12.39 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	115.60	207.00	154.60 207.00
Radial Flange	14.46	138.00	19.34 138.00
Tangential Flange	82.10	138.00	109.79 138.00
Maximum Average	98.85	138.00	132.20 138.00
Bolting	134.10	159.00	12.39 159.00

Minimum Required Flange Thickness 293.243 mm.

Estimated M.A.W.P. (Operating) 152 bars

Estimated Finished Weight of Flange at given Thk. 2632.6 kg.

Estimated Unfinished Weight of Forging at given Thk 3233.5 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 = $52.14 \cdot 305979.9 / 1.0 \cdot 9806.637 \cdot 0.448 / (4.291 \cdot 201334 \cdot 56.825^2 \cdot 233.728 \cdot 0.3)$
 = 0.358 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 = $52.14 \cdot 228790.7 / 1.0 \cdot 9806.637 \cdot 0.448 / (4.291 \cdot 197675 \cdot 56.825^2 \cdot 233.728 \cdot 0.3)$
 = 0.272 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 = $52.14 \cdot 305979.9 / 1.0 \cdot 9806.637 \cdot 0.448 / (4.521 \cdot 201334 \cdot 56.825^2 \cdot 233.728 \cdot 0.3)$
 = 0.339 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 = $52.14 \cdot 228790.7 / 1.0 \cdot 9806.637 \cdot 0.448 / (4.521 \cdot 197675 \cdot 56.825^2 \cdot 233.728 \cdot 0.3)$
 = 0.258 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	115.86	207.00	153.34 207.00
Radial Flange	15.70	138.00	20.78 138.00
Tangential Flange	80.03	138.00	105.91 138.00
Maximum Average	97.95	138.00	129.62 138.00
Bolting	134.22	159.00	12.47 159.00



Minimum Req'd Thickness, ambient, no Corrosion: 289.890 mm.

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Flange Input Data Values

Description: CH. FLG-2 :

CH. FLG-2

Flange Type	Integral Weld Neck		
Design Pressure	P	147.09	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	955.000	mm.
Flange Outside Diameter	A	1508.000	mm.
Flange Thickness	t	295.0000	mm.
Thickness of Hub at Small End	go	60.0000	mm.
Thickness of Hub at Large End	g1	70.0000	mm.
Length of Hub	h	90.0000	mm.
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	1360.000	mm.
Nominal Bolt Diameter	a	80.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	990.000	mm.
Flange Face Inside Diameter	Fid	955.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	987.000	mm.
Gasket Inside Diameter	Gi	961.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62.06	N./mm ²
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (147.09 \cdot (955.0/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 147.09) + Ca$$

$$= 57.7200 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 57.0) / (480.5 + 0.6 \cdot 57.0)$$

$$= 152.815 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 60.0) / (477.5 + 0.6 \cdot 60.0)$$

$$= 161.234 \text{ bars}$$

Corroded Flange ID,	Bcor = B+2*ci	961.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	67.000 mm.

Corroded Small Hub, $g0Cor = go-ci$ 57.000 mm.
 Code R Dimension, $R = ((C-Bcor)/2)-g1cor$ 132.500 mm.
 Gasket Contact Width, $N = (Go - Gi)/2$ 13.000 mm.
 Basic Gasket Width, $bo = N/2$ 6.500 mm.
 Effective Gasket Width, $b = Cb*\sqrt{bo}$ 6.425 mm.
 Gasket Reaction Diameter, $G = Go - 2 * b$ 974.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 974.1509^2 \cdot 147.094$$

$$= 1117976.625 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 974.1509 \cdot 3.75 \cdot 147.09$$

$$= 221192.797 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 961.0^2 \cdot 147.0936$$

$$= 1087995.375 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 1117977 - 1087995$$

$$= 29981.258 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1117977 + 221193 + 0, 0)$$

$$= 1339169.500 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 62.06 \cdot 6.4246 \cdot \pi \cdot 974.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 124413.508 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((1339170 + 0 + 4 \cdot 0/974.15)/159, 124414/159)$$

$$= 825.990 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 80.0 + 6 \cdot 295.0/(3.75 + 0.5)$$

$$= 576.471 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1360.0 \cdot \sin(\pi/24)$$

$$= 177.516 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs/(2a + t) }, 1)$$

$$= \max(\sqrt{ 177.516/(2 \cdot 80.0 + 295.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	825.990	978.440	
Radial Distance between Hub and Bolts:	91.500	132.500	
Radial Distance between Bolts and Edge:	74.000	74.000	
Circ. Spacing between the Bolts:	162.600	177.516	576.471

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 978.44 \cdot 159.0 / (62.06 \cdot \pi (987.0 + 961.0))$$

$$= 40.964 \text{ mm. [Note: Exceeds actual gasket width, 13.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa (Am + Ab) / 2$$

$$= 159.0 (825.9903 + 978.4396) / 2$$

$$= 1462751.75 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1339170 - 1117977$$

$$= 221192.81 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1360.0 - 974.1509) / 2$$

$$= 192.9246 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1360.0 - 961.0) / 2 + 192.9246)$$

$$= 196.2123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1360.0 - 961.0 - 67.0) / 2$$

$$= 166.0000 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1087995.	166.0000	1.0000	180611.
Face Pressure, Mt	29981.	196.2123	1.0000	5883.
Gasket Load, Mg	221193.	192.9246	1.0000	42674.
Gasket Seating, Matm	1462752.	192.9246	1.0000	282206.

Total Moment for Operation, Mo 229168. Kg-m.
 Total Moment for Gasket seating, Mg 282206. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 234.045 mm.
 Hub Ratio, h/h0 = HL / H0 0.385
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.175

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.889
 Factor V per 4.16.5 0.448
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1508.0 / 961.0$$

$$= 1.569$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.569^2 \cdot \log_{10}(1.569))) / (1.569^2 - 1)) / (1.569 - 1)$$

$$= 4.484$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.569^2 (1 + 8.55246 \cdot \log_{10}(1.569)) - 1) / ((1.0472 + 1.9448 \cdot 1.569^2) \cdot (1.569 - 1))$$

= 1.681

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.569^2(1 + 8.55246 \cdot \log_{10}(1.569)) - 1) / (1.36136(1.569^2 - 1) \cdot (1.569 - 1))$$

$$= 4.927$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.569^2 + 1) / (1.569^2 - 1)$$

$$= 2.368$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 4.927 \cdot 57.0^2 \cdot 234.045 / 0.448$$

$$= 8364237.500 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.889 / 234.045$$

$$= 0.004 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / (T + t^3/d)$$

$$= (295.0 \cdot 0.004 + 1) / (1.681 + 295.0^3 / 8364238)$$

$$= 4.331$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 229168 / 961.0) / (4.3309 \cdot 67.0^2)$$

$$= 120.29 \text{ N/mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 282206 / 961.0) / (4.3309 \cdot 67.0^2)$$

$$= 148.13 \text{ N/mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.494 \cdot 229168 / 961.0) / (4.3309 \cdot 295.0^2)$$

$$= 15.47 \text{ N/mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.494 \cdot 282206 / 961.0) / (4.3309 \cdot 295.0^2)$$

$$= 19.06 \text{ N/mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.4836 \cdot 229168 / (295.0^2 \cdot 961.0)) - 2.3676 \cdot 15$$

$$= 83.85 \text{ N/mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.4836 \cdot 282206 / (295.0^2 \cdot 961.0)) - 2.3676 \cdot 19$$

$$= 103.25 \text{ N/mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (120 + \max(15, 84)) / 2$$

$$= 102.07 \text{ N/mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (148 + \max(19, 103)) / 2$$

$$= 125.69 \text{ N/mm}^2$$

Bolt Stress, Operating [BSo]:

= Wm1 / Ab
 = 1339170/978.4396
 = 134.22 N./mm²

Bolt Stress, Seating [BSa]:
 = (Wm2 / Ab)
 = (124414/978.4396)
 = 12.47 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	120.29	207.00	148.13	207.00
Radial Flange	15.47	138.00	19.06	138.00
Tangential Flange	83.85	138.00	103.25	138.00
Maximum Average	102.07	138.00	125.69	138.00
Bolting	134.22	159.00	12.47	159.00

Minimum Required Flange Thickness 280.391 mm.

Estimated M.A.W.P. (Operating) 153 bars
 Estimated Finished Weight of Flange at given Thk. 2591.1 kg.
 Estimated Unfinished Weight of Forging at given Thk 3192.1 kg.

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:
 = max(1; F1num / F1den)
 = max(1; 0.7888/1.4456)
 = 1.0000

Hub Stress Correction Factor [f2]:
 = max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 0.3274/1.5011) if h/ho < 0.35, else max (0.25; 0.1764/0.7678)

= 0.2500

In this case, the hub ratio h/h_o , 0.3845 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	S_{bmax}	507.0 N./mm ²
Minimum permissible bolt stress	S_{bmin}	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	S_{fmax}	436.0 N./mm ²
Target assembly gasket stress	S_gT	207.0 N./mm ²
Maximum permissible gasket stress	S_{gmax}	275.0 N./mm ²
Minimum gasket operating stress	S_{gmin-O}	97.0 N./mm ²
Minimum gasket seating stress	S_{gmin-S}	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	40.7683 cm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	S_{ya}	250.0 N./mm ²
Flange Material Operating Yield Stress	S_{yo}	229.6 N./mm ²

Gasket Contact Area [A_g]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (987.0^2 - 961.0^2)$$

$$= 397.8 \text{ cm}^2$$

Target Bolt Stress [S_{bsel}]: (Eq. O-1)

$$= S_gT \cdot A_g / (nb \cdot Abs)$$

$$= 207.0 \cdot 397.788 / (24 \cdot 40.768)$$

$$= 84.2 \text{ N./mm}^2$$

Upper Bolt Stress Limit [S_{bsel}]: (Eq. O-4)

$$= \min(S_{bsel}; S_{bmax})$$

$$= \min(84.157; 507.0)$$

$$= 84.2 \text{ N./mm}^2$$

Lower Bolt Stress Limit [S_{bsel}]: (Eq. O-5)

$$= \max(S_{bsel}; S_{bmin})$$

$$= \max(84.157; 290.0)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [S_{bsel}]: (Eq. O-6)

$$= \min(S_{bsel}; S_{fmax})$$

$$= \min(290.0; 436.0)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: S_{bsel} \geq S_{gmin-S} (A_g / (nb \cdot Abs))$$

$$:: 290.0 \geq 140.0 (397.788 / (24 \cdot 40.768))$$

$$:: 290.0 \geq 56.9 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: S_{bsel} \geq (S_{gmin-O} \cdot A_g + \pi/4 \cdot P_{max} \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290.0 \geq (97.0 \cdot 397.788 + \pi/4 \cdot 147.094 \cdot 961.0^2) / (0.7 \cdot 40.768 \cdot 24)$$

$$:: 290.0 \geq 212.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: S_{bsel} \leq S_{gmax} (A_g / (nb \cdot Abs))$$

$$:: 290.0 \leq 275.0 (397.788 / (24 \cdot 40.768))$$

$$:: 290.0 \leq 111.8 \text{ N./mm}^2 \text{ Check Fails}$$

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 282206.3 / 1.0 \cdot 9806.637 \cdot 0.448 / (3.864 \cdot 201334 \cdot 57.0^2 \cdot 234.045 \cdot 0.3)$$

$$= 0.364 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 229168.0 / 1.0 \cdot 9806.637 \cdot 0.448 / (3.864 \cdot 197675 \cdot 57.0^2 \cdot 234.045 \cdot 0.3)$$

$$= 0.301 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 282206.3 / 1.0 \cdot 9806.637 \cdot 0.448 / (4.331 \cdot 201334 \cdot 57.0^2 \cdot 234.045 \cdot 0.3)$$

$$= 0.324 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 229168.0 / 1.0 \cdot 9806.637 \cdot 0.448 / (4.331 \cdot 197675 \cdot 57.0^2 \cdot 234.045 \cdot 0.3)$$

$$= 0.268 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	120.31	207.00	146.82	207.00
Radial Flange	16.69	138.00	20.37	138.00
Tangential Flange	81.70	138.00	99.70	138.00
Maximum Average	101.01	138.00	123.26	138.00
Bolting	134.22	159.00	12.47	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 276.962 mm.

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Flange Input Data Values

Description: CH. COVER-2 :

CH. COVER-2

Flange Type		Blind	
Design Pressure	P	147.00	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1508.000	mm.
Flange Thickness	t	265.0000	mm.
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	4272.566	mm.
Diameter of Bolt Circle	C	1360.000	mm.
Nominal Bolt Diameter	a	80.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	990.000	mm.
Flange Face Inside Diameter	Fid	0.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	987.000	mm.
Gasket Inside Diameter	Gi	961.000	mm.
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.

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Gasket Contact Width,	$N = (Go - Gi)/2$	13.000	mm.
Basic Gasket Width,	$bo = N/2$	6.500	mm.
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425	mm.
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	974.151	mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 974.1509^2 \cdot 147.0$$

$$= 1117265.250 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 974.1509 \cdot 3.0 \cdot 147.0$$

$$= 176841.625 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\
 &= \max(1117265 + 176842 + 0, 0) \\
 &= 1294106.875 \text{ Kgf}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.95 \cdot 6.4246 \cdot \pi \cdot 974.151 + 0.0 \cdot 0.0 \cdot 0.0 \\
 &= 138237.203 \text{ Kgf}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((1294107 + 0 + 4 \cdot 0 / 974.15) / 159, 138237 / 159) \\
 &= 798.196 \text{ cm}^2
 \end{aligned}$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	798.196	978.440	
Radial Distance between Bolts and Edge:	74.000	74.000	
Circ. Spacing between the Bolts:	162.600	177.516	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 978.44 \cdot 159.0 / (68.95 \cdot \pi (987.0 + 961.0)) \\
 &= 36.868 \text{ mm. [Note: Exceeds actual gasket width, 13.000]}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a (A_m + A_b) / 2 \\
 &= 159.0 (798.196 + 978.4396) / 2 \\
 &= 1440220.38 \text{ Kgf}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o \\
 &= 1294106.88 \text{ Kgf}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (1360.0 - 974.1509) / 2 \\
 &= 192.9246 \text{ mm.}
 \end{aligned}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$\begin{aligned}
 &= 1.9 \cdot W_{m1} \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2 \\
 &= 1.9 \cdot 1294107 \cdot 192.9246 \cdot 1.0 / (262.0^2 \cdot 974.1509) + 0.3 \cdot 1.0 \cdot 147.0 \cdot 974.1509^2 / 262.0^2 \\
 &= 130.54 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$\begin{aligned}
 &= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G) \\
 &= 1.9 \cdot 1440220 \cdot 192.9246 \cdot 1.0 / (262.0^2 \cdot 974.1509) \\
 &= 77.42 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= W_{m1} / A_b \\
 &= 1294107 / 978.4396 \\
 &= 129.71 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (W_{m2} / A_b) \\
 &= (138237 / 978.4396) \\
 &= 13.86 \text{ N./mm}^2
 \end{aligned}$$

Flange Stress Analysis Results: N./mm²

Actual	Operating Allowed	Actual	Gasket Seating Allowed



Tangential Flange	130.54	138.00	77.42	138.00
Bolting	129.71	159.00	13.86	159.00

Reqd. Blind Flange Thickness at Center 257.820 mm.
 Reqd. Blind Flange Thickness at Gasket 199.246 mm.

Estimated M.A.W.P. (Operating) 155 bars
 Estimated Finished Weight of Flange at given Thk. 3668.3 kg.
 Estimated Unfinished Weight of Forging at given Thk 3668.3 kg.

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Tangential Flange	127.60	138.00	75.68	138.00
Bolting	129.71	159.00	13.86	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 254.838 mm.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
CH. COVER-1		147	...	3	1508	138	1
CH. FLG-1		147.09	...	3.175	955	138	2
CH. SHELL-1		147.09	60	3	955	138	3
MAIN SHELL		191.09	77	3	955	138	4
CH. SHELL-2		147.09	60	3	955	138	5
CH. FLG-2		147.09	...	3	955	138	6
CH. COVER-2		147	...	3	1508	138	7

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
CH. COVER-1		147	155.4	158.98	265	257.82	1
CH. FLG-1		147	152.232	161.234	300	293.243	2
CH. SHELL-1		147	152.721	161.234	60	57.7201	3
MAIN SHELL		191	194.443	202.887	77	75.5718	4
CH. SHELL-2		147	152.721	161.234	60	57.7201	5
CH. FLG-2		147	152.722	161.234	295	280.391	6
CH. COVER-2		147	155.4	158.98	265	257.82	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 192 bars
 Shell Side MAPnc = 199 bars
 Channel Side MAWP = 149 bars
 Channel Side MAPnc = 159 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: CH. SHELL-1

Cylindrical Shell Design Information:					
Design Pressure	P	147.094 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1200.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	955.000 mm.	Actual Stress		132.8 N./mm ²
Spec. Min. Thk t _e , or s _a		60.0000 mm.	Spec. Nominal Thk.		60.000 mm.
Surface Area		40526.6 cm ²	SG of Contents		1.000
Element Volume		859717.4 cm ³	Weight of Contents		859.2 Kg _f
Empty Weight		1779.4 Kg _f	Operating Weight		2638.6 Kg _f
			Comp. All. Ope. Stress		147.3 N./mm ²

Radiography Information:			
User Defined			User Defined
Circ. Joint Efficiency	100 %		Long. Joint Eff. E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (147.094 \cdot 480.5) / (138.0 \cdot 1.0 \cdot 0.6 \cdot 147.094)$$

$$= 54.7201 + 3.0000 = 57.7201 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

[Less Operating Hydrostatic Head Pressure of 0.094 bars](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 57.0) / (480.5 + 0.6 \cdot 57.0)$$

$$= 153 - 0.1 = 153 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 60.0) / (477.5 + 0.6 \cdot 60.0)$$

$$= 161 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (147.094 \cdot (480.5 + 0.6 \cdot 57.0)) / (1.0 \cdot 57.0)$$

$$= 132.831 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 5.911 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 60.0, t_r = 55.581, c = 3.0 mm., E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.975, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 19 °C

Note:

[Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.](#)

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	149.258 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	60.000 mm.	Basic MDMT	21 °C
Stress/Thk. Ratio	0.975	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	19 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: MAIN SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	191.094 bars	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		6840.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	955.000 mm.	Actual Stress		135.6 N./mm ²
Spec. Min. Thk t _e , or s _a		77.0000 mm.	Spec. Nominal Thk.		77.000 mm.
Surface Area		238307.4 cm ²	SG of Contents		1.000
Element Volume		3628183.5 cm ³	Weight of Contents		3626.0 Kg
Empty Weight		13234.5 Kg	Operating Weight		16860.4 Kg
			Comp. All. Ope. Stress		141.3 N./mm ²

Radiography Information:			
User Defined			User Defined
Circ. Joint Efficiency	100 %		Long. Joint Eff. E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (191.094 \cdot 480.5) / (138.0 \cdot 1.0 \cdot 0.6 \cdot 191.094)$$

$$= 72.5718 + 3.0000 = 75.5718 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.094 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 74.0) / (480.5 + 0.6 \cdot 74.0)$$

$$= 195 - 0.1 = 194 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 77.0) / (477.5 + 0.6 \cdot 77.0)$$

$$= 203 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (191.094 \cdot (480.5 + 0.6 \cdot 74.0)) / (1.0 \cdot 74.0)$$

$$= 135.555 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 7.461 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Govern. thk, t_g = 77.0, t_r = 72.936, c = 3.0 mm., E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.986, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 25 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	191.972 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	77.000 mm.	Basic MDMT	26 °C
Stress/Thk. Ratio	0.986	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	25 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: CH. SHELL-2

Cylindrical Shell Design Information:					
Design Pressure	P	147.094 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1200.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	955.000 mm.	Actual Stress		132.8 N./mm ²
Spec. Min. Thk t _e , or s _a		60.0000 mm.	Spec. Nominal Thk.		60.000 mm.
Surface Area		40526.6 cm ²	SG of Contents		1.000
Element Volume		859717.4 cm ³	Weight of Contents		859.2 Kg _f
Empty Weight		1779.4 Kg _f	Operating Weight		2638.6 Kg _f
			Comp. All. Ope. Stress		141.3 N./mm ²

Radiography Information:			
User Defined			User Defined
Circ. Joint Efficiency	100 %		Long. Joint Eff. E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (147.094 \cdot 480.5) / (138.0 \cdot 1.0 - 0.6 \cdot 147.094)$$

$$= 54.7201 + 3.0000 = 57.7201 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.094 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 57.0) / (480.5 + 0.6 \cdot 57.0)$$

$$= 153 - 0.1 = 153 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 60.0) / (477.5 + 0.6 \cdot 60.0)$$

$$= 161 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (147.094 \cdot (480.5 + 0.6 \cdot 57.0)) / (1.0 \cdot 57.0)$$

$$= 132.831 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 5.911 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 60.0, t_r = 55.581, c = 3.0 mm., E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.975, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 19 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:	
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MDMT Curve	B	Press. at MDMT	149.258 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	60.000 mm.	Basic MDMT	21 °C
Stress/Thk. Ratio	0.975	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	19 °C
		Joint Efficiency E*	100 %

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	249.563 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	258.520 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	211.169 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	191.972 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	194.035 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	191.100 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	206.526 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	164.184 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	210.210 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	210.210 bars
App 27-4	= M.A.W.P.	149.258 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
CH. SHELL-1	175.3	234.0	0.749	194.13
MAIN SHELL	177.1	234.0	0.757	249.66
CH. SHELL-2	175.3	234.0	0.749	194.13

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
TS OUT NOZ. T2	Nozzle	138.00	138.00	1.000
DRAIN NOZ. D1	Nozzle	138.00	138.00	1.000
SH. IN NOZ. S1	Nozzle	138.00	138.00	1.000
SH. OUT NOZ. S2	Nozzle	138.00	138.00	1.000
TS. IN NOZ. T1	Nozzle	138.00	138.00	1.000
VENT NOZ. V1	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
CH. COVER-1	138.00	138.00	1.000
CH. FLG-1	138.00	138.00	1.000
CH. SHELL-1	138.00	138.00	1.000
MAIN SHELL	138.00	138.00	1.000
CH. SHELL-2	138.00	138.00	1.000
CH. FLG-2	138.00	138.00	1.000
CH. COVER-2	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000



Tubesheet Material	138.00	138.00	1.000
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Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
TS OUT NOZ. T2	118.43	225.00	0.526
DRAIN NOZ. D1	94.29	225.00	0.419
SH. IN NOZ. S1	149.80	225.00	0.666
SH. OUT NOZ. S2	150.90	225.00	0.671
TS. IN NOZ. T1	118.43	225.00	0.526
VENT NOZ. V1	94.29	225.00	0.419

Elements Suitable for Test Pressure.

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MDMT Calculations for Stiffening Rings:

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Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: MAIN SHELL

Shell Test Pressure		249.56 bars
Shell Thickness	ts	77.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	955.000 mm.
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH. SHELL-1

Channel Type:		Cylinder
Channel Test Pressure		194.04 bars
Channel Thickness	tc	60.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	955.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	367
Tube Wall Thickness	et	3.4100 mm.
Tube Outside Diameter	D	25.4000 mm.
Total Straight Tube Length	Lt	7200.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		6840.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	39.3700 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm.
Groove Weld Leg	ag	3.0000 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	438.406 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	4389.4 cm ²
Perimeter of Tube Layout	Cp	3589.6 mm.

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 180.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm.
 Tubesheet Outside Diameter A 1109.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 1608.000 mm.
 Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
 Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (194.04 \cdot (25.4/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 194.04)$
 $= 2.4605 + 0.0000 = 2.4605 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
3.4100	25.40	6840.00	7.45	50.0000	0.0198260	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 198.9763 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
4.0291	25.40	6840.00	6.30	50.0000	0.0276781	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 249.5931 \text{ bars}$

Summary of Tube Required Thickness Results:

<i>Total Required Thickness including Corrosion all.</i>	<i>4.0291 mm.</i>
Allowable Internal Pressure at Corroded thickness	277.93 bars
Required Internal Design Pressure	194.04 bars
<i>Allowable External Pressure at Corroded thickness</i>	<i>198.98 bars</i>
Required External Design Pressure	249.56 bars
Required Thickness due to Shell Side pressure	4.0291 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$= l_{tx} / h$ (modified for corrosion if present)
 $= 0/180.0$
 $= 0.0000$ (must be $0 \leq p \leq 1$)

Effective tube hole diameter [d*]:

$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$
 $= \max(25.4 - 2 \cdot 3.41 (0.20270E+09/0.20270E+09) \cdot$
 $(92/137) \cdot (0); 25.4 - 2 \cdot 3.41)$
 $= 25.4000 \text{ mm.}$

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot ro + dt$
 $= 2 \cdot 438.406 + 25.4$
 $= 902.212 \text{ mm.}$

Determine the basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (39.37 - 25.4) / 39.37$
 $= 0.355$

Equivalent outer tube limit radius [ao]:

$= Do / 2$
 $= 902.212/2$

= 451.1060 mm.

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$$

$$= 39.37 / \sqrt{(1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 902.212 \cdot 39.37) / (\pi \cdot 902.212^2))}$$

$$= 39.3700 \text{ mm.}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (39.37 - 25.4) / 39.37$$

$$= 0.3548$$

Ratio [ps]:

$$= as / ao$$

$$= 477.5 / 451.106$$

$$= 1.0585$$

Ratio [pc]:

$$= ac / ao$$

$$= 477.5 / 451.106$$

$$= 1.0585$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 367 ((25.4 - 2 \cdot 3.41) / (2 \cdot 451.106))^2$$

$$= 0.8444$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 367 (25.4 / (2 \cdot 451.106))^2$$

$$= 0.7091$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts)Es/L$$

$$= \pi \cdot 77.0 (955.0 + 77.0).20270E+09/6840.0$$

$$= 754426.7500 \text{ Kgf/mm.}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt)Et/L$$

$$= \pi \cdot 3.41 (25.4 - 3.41).20270E+09/6840.0$$

$$= 711.9119 \text{ Kgf/mm.}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 754426 / (367 \cdot 711.912)$$

$$= 2.88752$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kj)$$

$$= 1 / (1 + 754426/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((955.0 + 77.0) 77.0)$$

$$= 0.0064 \text{ 1/mm.}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.006 \cdot 0.20270E+09 \cdot 77.0^3 / (6(1 - 0.3^2))$$

$$= 1092965248.0000 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient [λ_s]:

$$\begin{aligned}
 &= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\
 &= 6(955.0 + 77.0) \cdot 0.10930E+1 / 180.0^3 \cdot (1 + 180.0 \cdot 0.006 + (180.0 \cdot 0.006)^2 / 2) \\
 &= 3289181.2500 \text{ bars}
 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned}
 &= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s))) \\
 &= 955.0^2 / (4 \cdot 0.20270E+09 \cdot 77.0) \cdot (1 - 955.0 \cdot 0.3 / (2(955.0 + 77.0))) \\
 &= 0.12580E-01 \text{ mm./N./mm}^2
 \end{aligned}$$

Channel coefficient [β_c]:

$$\begin{aligned}
 &= ((12(1 - v_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((955.0 + 60.0) 60.0) } \\
 &= 0.0074 \text{ 1/mm.}
 \end{aligned}$$

Channel coefficient [k_c]:

$$\begin{aligned}
 &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_c^2)) \\
 &= 0.007 \cdot 0.20270E+09 \cdot 60.0^3 / (6(1 - 0.3^2)) \\
 &= 590696576.0000 \text{ bars} \cdot \text{mm.}^2
 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$\begin{aligned}
 &= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\
 &= 6(955.0 + 60.0 / \cos(0)) \cdot 0.59070E+09 / 180.0^3 \cdot (1.0 + 180.0 \cdot 0.007 \cdot (1.0 + 1.0) / 2 + (180.0 \cdot 0.007)^2 / 2 \cdot 1.0) \\
 &= 1976914.5 \text{ bars}
 \end{aligned}$$

Channel coefficient [δ_c]:

$$\begin{aligned}
 &= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c))) \\
 &= 955.0^2 / (4 \cdot 0.20270E+09 \cdot 60.0) \cdot (1 - 955.0 \cdot 0.3 / (2(955.0 + 60.0))) \\
 &= 0.16101E-01 \text{ mm./N./mm}^2
 \end{aligned}$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$\begin{aligned}
 h/p &= 4.572009 ; \mu^* = 0.354839 ; E^* = 71672888 \text{ KPa.} \\
 E^*/E &= 0.353589 ; v^* = 0.324557
 \end{aligned}$$

Note: As h/p (4.572) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$\begin{aligned}
 &= ((24(1 - v^{*2}) Nt \cdot Et \cdot t(dt - tt) a_o^2) / (E^* \cdot L \cdot H^3))^{1/4} \\
 &= ((24(1 - 0.325^2) 367 \cdot 0.20270E+09 \cdot 3.41 (25.4 - 3.41) 451.106^2) / (71672888 \cdot 6840.0 \cdot 180.0^3))^{1/4} \\
 &= 1.7087
 \end{aligned}$$

Values from Table 4.18.3

$Z_d =$	0.287102 ;	$Z_v =$	0.173474 ;	$Z_m =$	0.709056
$Z_a =$	0.629339E+00 ;	$Z_w =$	0.173474		

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / D_o \\
 &= 1109.0 / 902.212 \\
 &= 1.2292
 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned}
 &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\
 &= (1 - 0.32) / 71672888 \cdot (3289181 + 1976914 + 0.20270E+09 \cdot \ln(1.23)) \\
 &= 5.3570
 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned}
 &= (1 + v^*) F \\
 &= (1 + 0.3246) 5.357
 \end{aligned}$$

= 7.0956

Compute parameter [Q1]:

$$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.0585 - 1 - 7.0956 \cdot 0.1735) / (1 + 7.0956 \cdot 0.7091)$$

$$= -0.194389$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.2871 + -0.19439 \cdot 0.17347) / 2 \cdot 1.70874^4$$

$$= 1.0801$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.17347 + -0.19439 \cdot 0.70906) / 2 \cdot 1.70874^4$$

$$= 0.1519$$

Compute parameter [U]:

$$= (Z_w + (p_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.1735 + (1.0585 - 1) 0.7091) 1.70874^4 / (1 + 7.0956 \cdot 0.7091)$$

$$= 0.3038$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

= 0 (config a)

Compute Parameter [y]:

= 0.000 mm. (For Pressure only cases)

Calculate parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.0585 \cdot 0.10930E+1 \cdot 0.0064 \cdot 0.01258 (1 + 180.0 \cdot 0.0064)$$

$$= 20280.4668 \text{ mm.}^2$$

Calculate parameter [w*s]:

$$= a o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - w_s$$

$$= 451.106^2 (1.059^2 - 1) \cdot (1.059 - 1) / 4 - 20280.467$$

$$= -19921.9551 \text{ mm.}^2$$

Parameter [wc]:

$$= p_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.059 \cdot 0.59070E+09 [0.016101 (1.0 + 0.007 \cdot 1.0 \cdot 180.0) - 0 (1.0/180.0 + 1.0 \cdot 0.007/2)]$$

$$= 17249.1328 \text{ mm.}^2$$

Parameter [w*c]:

$$= a o^2 [(p_c^2 + 1) \cdot (p_c - 1) / 4 - (p_s - 1) / 2] - w_c$$

$$= 451.10599^2 [(1.05851^2 + 1) \cdot (1.05851 - 1) / 4 - (1.05851 - 1) / 2] - 17249.13281$$

$$= -16890.6211 \text{ mm.}^2$$

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(D_j^2 - (D_s)^2) / D_o^2] \}$$

$$= 249.563 \{ 0.709 + 2(1 - 0.709) 0.3 + [2/2.888 \cdot (955.0/902.212)^2] 0.3 - [(1.059^2 - 1) / (1.0 \cdot 2.888)] - [(1 - 1.0) / (2 \cdot 1.0 \cdot 2.888)] [(0^2 - (955.0)^2) / 902.212^2] \}$$

$$= 268.2191 \text{ bars}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.844 + 2(1 - 0.844) 0.3 + 1/(1. \cdot 2.888)] 194.035$$

$$= 249.1530 \text{ bars}$$

Pressure [P_y]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 367 \cdot 711.912 \cdot 0 / (3.142 \cdot 451.106^2)$$

$$= 0.0000 \text{ bars}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 0.304 \cdot 0 / (2 \cdot \pi \cdot 451.106^2)$$

$$= -0.0000 \text{ bars}$$

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t)$$

$$= -(0.304/451.106^2) \cdot (-19921.955 \cdot 249.563 - -16890.621 \cdot 194.035)$$

$$= 2.5300 \text{ bars}$$

Calculate the pressure [P_w]:

$$= (U/a_o^2) \cdot (\omega_s \cdot P^* s - \omega_c \cdot P^* c)$$

$$= (0.304/451.106^2) \cdot (20280.4668 \cdot 0 - 17249.1328 \cdot 0)$$

$$= 0.0000 \text{ bars}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1) Qz2)) \cdot$$

$$(P^* s - P^* t + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 2.888 / (1 + 1.0 \cdot 2.888 (1.08 + (1.059 -$$

$$1) 0.152)) \cdot (268.219 - 249.153 + 0 + 0 + 0 + 2.53)$$

$$= 15.0469 \text{ bars}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega^* s \cdot P_s - \omega^* c \cdot P_t) - (\omega_s \cdot P^* s - \omega_c \cdot P^* c)) +$$

$$W^* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-19921.955 \cdot 249.563 - -16890.621 \cdot 194.035) -$$

$$(20280.467 \cdot 0 - 17249.133 \cdot 0)) \cdot 0.01 +$$

$$0 \cdot 0 / (2 \cdot \pi)] / (1 + 7.0956 \cdot 0.70906)$$

$$= -2864.9282 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2)$$

$$= -0.194 + 2 \cdot -2864.928 / (15.047 \cdot 451.106^2)$$

$$= -0.37789$$

Fm Value from Table 4.18.3 = 0.188946

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu^*) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.1889 / 0.3548) \cdot (2 \cdot 451.106 / (180.0 - 0))^2 \cdot 15.05$$

$$= 30.1955 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 250.0 = 356.24 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu^*) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.023 / 0.3548) \cdot (2 \cdot 451.106 / (8.59 - 0))^2 \cdot 3.32$$

$$= 356.2255 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 8.5899 + 0 + 0$$

$$= 8.5899 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot Sy)$$

$$= (2 \cdot 0.355 \cdot 180.0 / 451.106) \cdot 189.997$$

$$= 53.8023 \text{ N./mm}^2$$

Shear Stress is not required to be computed; $[Pe] \leq \tau_{\text{limit}}$

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\begin{aligned} \text{Tubesheet thickness (Incl. Corr.)} &= 3.8100 \text{ mm.} \\ \text{Tubesheet Shear Stress} &= 16.9992 \text{ N./mm}^2 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) $[H_{\text{reqd}}]$:

$$\begin{aligned} &= \max(H_{\text{reqB}}; H_{\text{reqS}}) \\ &= \max(8.5899; 3.81) \\ &= 8.5899 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{\text{tmin}} = 0.9444, F_{\text{tmax}} = 1.0194$$

First Extreme Tube Axial Stress from among all the tubes $[\sigma_1]$:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{\text{tmin}}) / (X_t - X_s) \\ &= ((249.56 \cdot 0.7091 - 194.04 \cdot 0.8444) - (15.047) \cdot 0.944) / \\ &\quad (0.8444 - 0.7091) \\ &= -0.7943 \text{ N./mm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes $[\sigma_2]$:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{\text{tmax}}) / (X_t - X_s) \\ &= ((249.56 \cdot 0.7091 - 194.04 \cdot 0.8444) - (15.047) \cdot 1.019) / \\ &\quad (0.8444 - 0.7091) \\ &= -1.6289 \text{ N./mm}^2 \end{aligned}$$

Maximum Tube Axial Stress $[\sigma_{\text{t,max}}]$:

$$= \max(|\sigma_{\text{t},1}|; |\sigma_{\text{t},2}|) = 1.629 \text{ N./mm}^2$$

Allowable Tube Stress $[\sigma_{\text{t,A}}]$:

$$= \beta_T \cdot S_{y,t} = 170.9969 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety $[F_s]$:

$$\begin{aligned} &= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25) \\ &= \max((3.25 - 0.25(0.287 + -0.378 \cdot 0.173)1.709^4); 1.25) \\ &= 2.0000 \text{ (Should be } \leq 2) \end{aligned}$$

Determine the factor $[r_t]$:

$$\begin{aligned} &= (\sqrt{d_t^2 + (d_t - 2 \cdot t_t)^2}) / 4 \\ &= (\sqrt{25.4^2 + (25.4 - 2 \cdot 3.41)^2}) / 4 \\ &= 7.8676 \text{ mm.} \end{aligned}$$

Determine the factor $[C_t]$:

$$\begin{aligned} &= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{y,t})} \\ &= \sqrt{(2 \cdot \pi^2 \cdot 0.20270\text{E}+09 / 179)} \\ &= 157.1179 \end{aligned}$$

Determine the factor $[F_t]$:

$$\begin{aligned} &= k \cdot L / r \\ &= 1.0 \cdot 1608.0 / 7.868 \\ &= 204.3835 \end{aligned}$$

Buckling Allowable Stress $[S_{\text{tb}}]$:

$$\begin{aligned} &= (1 / F_s) \cdot (\pi^2 \cdot E_t / F_t^2) (C_t \leq F_t) \\ &= (1 / 2.0) \cdot (\pi^2 \cdot 0.20270\text{E}+09 / 204.3835^2) \\ &= 170.997 \text{ N./mm}^2 \text{ (Never greater than } S_{\text{ot}}) \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load $[W_t]$:

$$\begin{aligned} &= \sigma_{\text{t,max}} \cdot \text{Tube Area} \\ &= 1.63 \cdot 2.3558 \\ &= 39.13 \text{ Kg} \end{aligned}$$

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (25.4 - 3.41) 3.41$
 $= 2.356 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 3.41 (25.4 - 3.41) 170.997$
 $= 4107.609 \text{ Kgf}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$
 $= 0.85 \cdot \pi \cdot 3.0 (25.4 + 0.67 \cdot 3.0) 171.0$
 $= 3828.7654 \text{ Kgf}$

Weld Strength Factor [fw]:
 $= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$
 $= 170.997 / 170.997$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$
 $= 0.55 \cdot \pi \cdot 3.0 (25.4 + 0.67 \cdot 3.0) 171.0$
 $= 2477.4363 \text{ Kgf}$

Minimum Length of Weld [ar]:
 $= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$
 $= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 3.41 \cdot (25.4 - 3.41) 1.0 \cdot 1.0)} - 0.75 \cdot 25.4 \right)$
 $= 4.002 \text{ mm.}$

Minimum Required Fillet Weld Leg [afr]: 2.0460 mm.
 Minimum Required Groove Weld Leg [agr]: 2.0460 mm.

Tube-Tubesheet Jt allowable, 4107.61 is $\geq$ tube strength 4107.61 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= \beta T \cdot S_{y,s} = 0.95 \cdot 260.0 = 370.49 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 451.106^2 / ((955.0 + 77.0) 77.0) \cdot [15.05 + (1.059^2 - 1)$
 $(249.56 - 194.04)] + 477.5^2 \cdot 194.04 / ((955.0 + 77.0) 77.0)$
 $= 61.2442 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 77.0 mm., the minimum Shell length adjacent to the tubesheet should be 488.112 mm.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	30.2 $\leq$	356.2 N./mm ²	Pass
Tubesheet Shear Stress	2.9 $\leq$	190.0 N./mm ²	Pass
Maximum Tube Stress	1.6 $\leq$	171.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-1.6 $\leq$	-171.0 N./mm ²	Pass

Maximum Force on any one Tube 39.1 <= 4107.6 Kgf Pass
 Axial Membrane Stress in Shell 61.2 <= 370.5 N./mm² Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	8.590	180.000	Ok
Tube-Tubesheet Fillet Weld Leg :	2.046	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	2.046	3.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses			Case Type	Pass/ Fail
		Bend	Allwd	Shear	Allwd		
D1uc	135.426	...	211	356	34	190	Ps+Pt-Th D1 Ok
D2uc	144.716	...	241	356	37	190	Ps+Pt-Th D2 Ok
D3uc	8.590	...	30	356	3	190	Ps+Pt-Th D3 Ok
Max:	144.7159	... mm.	0.677		0.194	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	----	Shell Stresses			:	----	Shell Band Stress			----	Pass Fail
	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	
D1uc	4	370	...	...	:	...	...	...	...	:	Ok
D2uc	57	370	...	...	:	...	...	...	...	:	Ok
D3uc	61	370	...	...	:	...	...	...	...	:	Ok
Max RATIO		0.165		...	:		...		...	:	

Tube, Shell and Channel Stress Summary:

Case#	----	Tube Stresses			Tube Loads		Shell Stress	Channel Stress	Stress	Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Allwd	Allwd	Allwd	
D1uc	17	170	-3	-170	415	4107	...	370	...	370 Ok
D2uc	18	170	-18	-170	435	4107	...	370	...	370 Ok
D3uc	2	170	-2	-170	39	4107	...	370	...	370 Ok
Max RATIO		0.106		0.106		0.106		...		...

Summary of Thickness Comparisons for 3 Load Cases:



Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	144.716	180.000	Ok
<i>Tube thickness :</i>	<i>4.029</i>	<i>3.410</i>	<i>Fail</i>
Tube-tubesheet fillet weld leg :	2.046	3.000	Ok
Tube-tubesheet groove weld leg :	2.046	3.000	Ok

Min Shell length of thk, (77.000) adj. to tubesheet: 488.112 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm.
 Due to Pressure Shell Compresses by : -0.870 mm.
 Due to Pressure + Thermal Shell Compresses by : -0.870 mm.

Summary of Exchanger Pressure and Thermal Displacements: (mm.)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.1202	...	...	...	...	...
2	-0.8705	...	...	...	...	...
3	-0.7502	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	...	262	No Calc	No Calc
20	30	No Calc	...	296.825	No Calc	No Calc
30	Tsht	1199.97	1075	57	0.015305	122.731
Tsht	40	No Calc	...	57	No Calc	No Calc
40	Tsht	6839.99	1109	74	0.0051628	119.758
Tsht	50	No Calc	...	74	No Calc	No Calc
50	60	1200.04	1075	57	0.015304	122.731
60	70	No Calc	...	292	No Calc	No Calc
70	80	No Calc	...	262	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	265	154.849	...	No Calc
20	30	300	293.243	...	No Calc
30	Tsht	60	No Calc	...	86.7629
Tsht	40	60	No Calc	...	No Calc
40	Tsht	77	No Calc	...	106.542
Tsht	50	77	No Calc	...	No Calc
50	60	60	No Calc	...	86.7629
60	70	295	199.949	...	No Calc
70	80	265	154.849	...	No Calc

Minimum

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External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1199.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	6839.99	No Calc	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	1200.04	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	3668.33	...	3668.33	...	366.833
20	30	2632.6	282276	2603.71	283136	263.26
30	40	1779.42	859717	1695.45	870554	177.942
40	50	13234.5	3628183	12755.8	3689952	1323.45
50	60	1779.42	859717	1695.45	870554	177.942
60	70	2591.15	278490	2564.21	279303	259.115
70	80	3668.33	...	3668.33	...	366.833
Total		29353	5908384.50	28651	5993500.00	2935



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
20	Liqd	282.104	195	...	...	Liquid: 20
30	Liqd	859.193	600	...	...	Liquid: 30
30	Nozl	983.335	500	680.7	...	TS OUT NOZ. T2
30	Nozl	29.8804	600	-498.925	...	DRAIN NOZ. D1
40	Sadl	338.228	1370	891.25	...	FIXED SADDLE
40	Sadl	338.228	5190	891.25	...	SLIDING SADDLE
40	Liqd	3625.97	3420	...	...	Liquid: 40
40	Insl	189.677	3417	...	...	HOT
40	Nozl	2486.22	800	782.3	...	SH. IN NOZ. S1
40	Nozl	2486	6034	-782.3	...	SH. OUT NOZ. S2
50	Liqd	859.193	600	...	...	Liquid: 50
50	Nozl	990.523	700	-680.7	...	TS. IN NOZ. T1
50	Nozl	17.9887	600	498.925	...	VENT NOZ. V1
60	Liqd	278.32	192.5	...	...	Liquid: 60
30	FTsh	1196.96	1290	...	...	Fixed Tubesheet
30	Tube	5307.02	4800	...	...	Tubes
30	RTsh	1196.96	8310	...	...	Tubesheet
40	Bafl	26.8009	960	...	...	Baffles
40	Bafl	26.8009	1318	...	...	Baffles
40	Bafl	26.8009	1676	...	...	Baffles
40	Bafl	26.8009	2034	...	...	Baffles
40	Bafl	26.8009	2392	...	...	Baffles
40	Bafl	26.8009	2750	...	...	Baffles
40	Bafl	26.8009	3108	...	...	Baffles
40	Bafl	26.8009	3466	...	...	Baffles
40	Bafl	26.8009	3824	...	...	Baffles
40	Bafl	26.8009	4182	...	...	Baffles
40	Bafl	26.8009	4540	...	...	Baffles
40	Bafl	26.8009	4898	...	...	Baffles
40	Bafl	26.8009	5256	...	...	Baffles
40	Bafl	26.8009	5614	...	...	Baffles
40	Bafl	26.8009	5972	...	...	Baffles
40	Bafl	26.8009	6330	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	676.5
Liquid	5904.8
Insulation	189.7
Nozzles	6993.9
Exchanger Components	8129.8



Liquid in Tubes 712.2
 Sum of the Detail Weights 22606.9 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	32289.1	32289.1	32289.1	32289.1	32289.1	32289.1
Saddles	676.5	676.5	676.5	676.5	676.5	676.5
Nozzles	6993.9	6993.9	6993.9	6993.9	6993.9	6993.9
Exchanger	8129.8	8129.8	8129.8	8129.8	8129.8	8129.8
Insulation	...	...	189.7	189.7	189.7	189.7
Ope. Liquid	...	...	...	...	...	5904.8
Tube Ope Lqd	...	...	...	...	...	712.2
Test Liquid	...	5904.8	...	...	...	...
Tube Tst Lqd	...	716.1	...	...	...	...
Totals	48089.3	54710.2	48278.9	48278.9	48278.9	54896.0

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	48089.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	54710.2 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	48278.9 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	48278.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	48278.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	54896.0 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	54123.3 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	716568.4 cm ³
Weight of Ope Liq in tubes :	712.2 kg.
Weight of Water in tubes :	716.1 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	30414.9	...
20	30	28006	11700.9
30	40	40526.6	36002.7
40	50	238307	205215
50	60	40526.6	36002.7
60	70	27769.2	11550.9
70	80	30414.9	...

Total	435965.5	300472.2 cm ²
Total	469.3	323.4 ft ²

Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
TS OUT NOZ. T2	238.12	255.10	85	1500	GR 1.1	127.787	81.773	38.123	29.397
DRAIN NOZ. D1	238.12	255.10	85	1500	GR 1.1	...	...	...	...
SH. IN NOZ. S1	219.10	255.10	200	1500	GR 1.1	85.192	99.640	19.101	19.101
SH. OUT NOZ. S	219.10	255.10	200	1500	GR 1.1	84.052	00.779	19.101	19.101
TS. IN NOZ. T1	238.12	255.10	85	1500	GR 1.1	127.787	81.773	38.123	29.397
VENT NOZ. V1	238.12	255.10	85	1500	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 199.640 bars
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 255.100 bars

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 181.773 bars
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 255.100 bars

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE KgF	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
TS OUT NOZ. T2	431.800	0.30	19191	13053	149.258	181.773	Pass
SH. IN NOZ. S1	647.700	0.30	43180	19580	191.972	199.640	Pass
SH. OUT NOZ. S2	647.700	0.30	42560	19580	191.972	200.779	Pass
TS. IN NOZ. T1	431.800	0.30	19191	13053	149.258	181.773	Pass

Evaluating Flange on Nozzle [TS OUT NOZ. T2]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(19191) + 4 \cdot \text{abs}(13053) \cdot 431.800 \leq \pi \cdot 431.800^3 \cdot ((238.123 - 149.258) + .300 \cdot 238.123)$$

$$= 0.32960E+06 \leq 0.41347E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle TS OUT NOZ. T2 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(238.123, 0.300 \cdot 238.123 + 238.123 - 329602.2 / (\pi \cdot 431.800^3))$$

$$= \min(238.123, 181.773)$$

$$= 181.773 \text{ bars}$$

Evaluating Flange on Nozzle [SH. IN NOZ. S1]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(43180) + 4 \cdot \text{abs}(19580) \cdot 647.700 \leq \pi \cdot 647.700^3 \cdot ((219.101 - 191.972) + .300 \cdot 219.101)$$

$$= 0.74161E+06 \leq 0.80836E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle SH. IN NOZ. S1 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(219.101, 0.300 \cdot 219.101 + 219.101 - 741606.8 / (\pi \cdot 647.700^3))$$

$$= \min(219.101, 199.640)$$

$$= 199.640 \text{ bars}$$

Evaluating Flange on Nozzle [SH. OUT NOZ. S2]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(42560) + 4 \cdot \text{abs}(19580) \cdot 647.700 \leq \pi \cdot 647.700^3 \cdot ((219.101 - 191.972) + .300 \cdot 219.101)$$

$$= 0.73169E+06 \leq 0.80836E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle SH. OUT NOZ. S2 [Pmax]:
= $\min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
= $\min(219.101, 0.300 \cdot 219.101 + 219.101 - 731689.5 / (\pi \cdot 647.700^3))$
= $\min(219.101, 200.779)$
= 200.779 bars

Evaluating Flange on Nozzle [TS. IN NOZ. T1]:
= $16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
= $16(19191) + 4 \cdot \text{abs}(13053) 431.800 \leq \pi \cdot 431.800^3 [(238.123 - 149.258) + 0.300 \cdot 238.123]$
= $0.32960E+06 \leq 0.41347E+06$ Check Passes

Maximum Pressure for Flange on Nozzle TS. IN NOZ. T1 [Pmax]:
= $\min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
= $\min(238.123, 0.300 \cdot 238.123 + 238.123 - 329602.2 / (\pi \cdot 431.800^3))$
= $\min(238.123, 181.773)$
= 181.773 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.986) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$
 $= \min(0.85, 0.918)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.615
Structure Height to Diameter ratio	9.720

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$; $Zg = 589788$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1400.0 + 300.0 = 1700.0$ mm.
 $= 5.577$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (5.577 ft.) < 15 ft.
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:
 $K1 = 0$, $K2 = 0$, $K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.615 \cdot 5.162$$

$$= 89.5 \text{ lbs. } [40.6] \text{ Kg}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
CH. COVER-1	1700.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. FLG-1	1700.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. SHELL-1	1700.0	0.000	0.000	0.000	1.035	1.000	161.849
MAIN SHELL	1700.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. SHELL-2	1700.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. FLG-2	1700.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. COVER-2	1700.0	0.000	0.000	0.000	1.035	1.000	161.849

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1700	1809.6	4795.44	161.849	44.6381
20	30	1700	1146	4469.4	161.849	41.6032
30	40	1700	1290	15480	161.849	144.095
40	50	1700	1426.72	97587.4	161.849	908.387
50	60	1700	1290	15480	161.849	144.095
60	70	1700	1146	4412.1	161.849	41.0698
70	80	1700	1809.6	4795.44	161.849	44.6381

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22
Importance Factor		1.500
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(7.06690.75)$
 = 0.087 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.087$, 0.03) per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.217 \cdot 53447 \cdot 1.5$
 = 5227.099 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 53447$
 = 2323.2 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	754	5938.54	812.542
20	30	477.5	5938.54	514.574
30	40	477.5	5938.54	514.574
40	Sad1	477.5	5938.54	514.574
Sad1	50	477.5	5938.54	514.574
40	50	477.5	5938.54	514.574



50	60	477.5	5938.54	514.574
60	70	477.5	5938.54	514.574
70	80	754	5938.54	812.542

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5318.175 mm.
Center of Gravity of the Liquid	5455.069 mm.
Center of Gravity of the Insulation	5455.175 mm.
Center of Gravity of the Nozzles	5453.238 mm.
Center of Gravity of the Tubesheet(s)	5458.175 mm.
Center of Gravity of the Tubes	5458.175 mm.
Center of Gravity of the Baffles	5683.174 mm.
Center of Gravity of Bare Shell New and Cold	5451.236 mm.
Center of Gravity of Bare Shell Corroded	5451.402 mm.
Vessel CG in the Operating Condition	5453.277 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	5452.844 mm.
Vessel CG in the Test Condition	5453.152 mm.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (250.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(160.0 + 1.56 \cdot \sqrt{ 517.5 \cdot 74.0 }, 2 \cdot 1370.0)$$

$$= 465.2781 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	517.50	mm.
Shell Thickness used in this Case	t	77.000	mm.
Stiffened Vessel Length per 4.15.6	L	6840.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1370.00	mm.
Saddle Width	b1 or b	160.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1400.00	mm.
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		37040.71	Kgf
Pressure used in Analysis	P	191.047	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	64.71	138.00
Long. Stress at Bottom of Midspan	68.90	138.00
Long. Stress at Top of Saddles	80.75	138.00
Long. Stress at Bottom of Saddles	59.08	138.00
Tangential Shear in Shell	6.66	110.40
Circ. Stress at Horn of Saddle	7.90	172.50
Circ. Compressive Stress in Shell	0.80	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (1368.5 / 2 + 0) \cdot 1400.0 / 991.5991$$

$$= 2898.3 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(247.64, 0) \cdot 1400.0 / 3820.0005$$

$$= 90.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(5227.1, 0) \cdot 1400.0 / 3820.0005$$

$$= 1915.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:



$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(5227/2 + 0) \cdot 1400.0/991.5991$$

$$= 11069.9 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 25971 + \max(91, 2898, 1916, 11070)$$

$$= 37040.7 \text{ Kg}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(0.0 + \pi/4(1.109 \cdot 1.2)^2)$$

$$= 2428.479 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	37378.94	Kgf
Transverse Shear Load	2613.55	Kgf
Longitudinal Shear Load	5227.10	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(955.0/2 + 77.0 + 18.0)\sin(120.0/2), 2 \cdot 517.5)$$

$$= 991.599 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -37041 \cdot 1370.0 [1 - (1 - 1370.0/6840.0 + (517.5^2 - 0.0^2)/(2 \cdot 1370.0 \cdot 6840.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6840.0))]$$

$$= -9439.1 \text{ Kg-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 37041 \cdot 6840/4((1 + 2(517.5^2 - 0^2)/(6840^2))/(1 + (4 \cdot 0)/(3 \cdot 6840)) - 4 \cdot 1370/6840)$$

$$= 13319.2 \text{ Kg-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 191.047 \cdot 517.5/(2 \cdot 74.0) - 13319.2/(\pi \cdot 517.5^2 \cdot 74.0)$$

$$= 64.71 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 191.047 \cdot 517.5/(2 \cdot 74.0) + 13319.2/(\pi \cdot 517.5^2 \cdot 74.0)$$

$$= 68.90 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 191.047 \cdot 517.5/(2 \cdot 74.0) - 9439.1/(0.1066 \cdot \pi \cdot 517.5^2 \cdot 74.0)$$

$$= 80.75 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 191.047 \cdot 517.5/(2 \cdot 74.0) + 9439.1/(0.1923 \cdot \pi \cdot 517.5^2 \cdot 74.0)$$

$$= 59.08 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 37041(6840.0 - 2 \cdot 1370.0)/(6840.0 + (4 \cdot 0.0/3))$$

$$= 22202.8 \text{ Kg}$$



Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 22202.77 / (517.5 \cdot 74.0)$$

$$= 6.66 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(517.5 \cdot 74.0)}$$

$$= 152.639 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 37041 \cdot 0.1 / (74.0(160.0 + 152.64 + 152.64))$$

$$= -0.80 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -37041 / (4 \cdot 74.0(160.0 + 152.639 + 152.639)) -$$

$$3 \cdot 0.0529 \cdot 37041 / (2 \cdot 74.0^2)$$

$$= -7.90 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(160.0 + 1.56 \cdot \sqrt{(517.5 \cdot 74.0)}, 2 \cdot 1370.0)$$

$$= 465.28 \text{ mm.}$$

Distance between Saddle Supports [Ls]:

$$= 3820.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000013 \cdot 3820.0(200.0 - 21.1)$$

$$= 8.713 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1090.0000	mm.
Baseplate Thickness	Bpthk	32.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	18.0000	mm.
Web Thickness	Webtk	18.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		200.0	°C
Saddle Yield Stress	Sy	225.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(480.5 \cdot 74.0)} + 250.0)^3) / 12$$

$$= 1837.562 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 1837.562 + 402.68(223.839 - 37.0)^2$$

$$= 142408.219 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	544.2	74.0	37.0	402.7	1489916.6	0.142E+06
Wearplate	250.0	18.0	83.0	45.0	373500.0	0.894E+04
Web	18.0	795.5	489.8	143.2	7012729.5	0.177E+06
BasePlate	200.0	32.0	903.5	64.0	5782399.5	0.296E+06
Totals	...	...	...	654.9	14658545.0	0.624E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14658545/654.87$$

$$= 223.839 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 37040.715$$

$$= 7538.5884 \text{ Kg}$$

Tension Stress, $St = (Fh/As) = 2.9315 \text{ N./mm}^2$

Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 134.9988 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1400.0 - 480.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 1002.630 \text{ mm.}$$

Bending Moment, $M = Fh \cdot d = 7558.5615 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 2.6598 \text{ N./mm}^2$

Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 149.9987 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(37041 + 338) \cdot 200.0 / (2 \cdot 1090.0 \cdot 150.0)}$$

$$= 25.935 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (477.5 + 77.0 + 18.0)$$

$$= 991.599 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 991.5991 / (3 - 1)$$

$$= 495.800 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 495.7996 \cdot 200.0/2$$

$$= 495.800 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 37378.938 / (1090.0 \cdot 200.0)$$

$$= 17.146 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 495.8 \cdot 17.15$$

$$= 8501.130 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 25.56 + 44.622$$

$$= 70.182 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

= 8501.1/70.182
 = 11.879 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	18.0	142.0	89.0	25.6	227484.0	0.109E+04
Web	247.9	18.0	9.0	44.6	40159.8	391.
Totals	...	...	...	70.2	267643.8	0.148E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 160.0 - 18.0
 = 142.000 mm.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 267643.75/70.182
 = 38.136 mm.

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(38.136, 160.0 - 38.136)
 = 121.864 mm.

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{1481.6/70.182}$
 = 45.947 mm.

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1400.0 - cos(120.0/2)(477.5 + 77.0 + 18.0) - 32.0
 = 1081.750 mm.

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 0.19258\text{E}+09/225.0}$
 = 129.985

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1081.75/45.947
 = 23.544

Bending Moment [Rm]:
 $= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
 $= 5227.1 / (2 \cdot 1090.0) \cdot 495.8 \cdot 1081.75/2$
 = 643.007 Kg-m.

Compressive Allowable, KL/r < Cc (23.5436 < 129.9855) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (23.54)^2 / (2 \cdot 129.99^2)) 225 / (5/3 + 3 \cdot (23.54) / (8 \cdot 129.99) - (23.54)^3 / (8 \cdot 129.99^3))$
 = 127.6 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
 = 11.88/127.64 + (643.01 • 121.864/14816142)/150.0
 = 0.439

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	18.0	142.0	89.0	25.6	227484.0	0.142E+04



Web	495.8	18.0	9.0	89.2	80319.5	307.
Totals	...	...	...	114.8	307803.5	0.173E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 307803.5/114.804$$

$$= 26.811 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(26.811, \text{abs}(160.0 - 26.811))$$

$$= 133.189 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1400.0 - \cos(572.5/2) (32.0 + 0.0 + 0.0) - 0.0$$

$$= 795.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1725.2/114.804) }$$

$$= 38.765 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 795.5/38.765$$

$$= 20.521$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 5227.1/(2 \cdot 1090.0)$$

$$= 2.398 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.398 \cdot 495.8 \cdot 795.5$$

$$= 945.713 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 945.713/129532.297$$

$$= 71.598 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (20.5209 < 129.9855) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (20.52)^2 / (2 \cdot 129.99^2)) 225 /$$

$$(5/3 + 3 \cdot (20.52) / (8 \cdot 129.99) - (20.52^3) / (8 \cdot 129.99^3))$$

$$= 128.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 14.2/128.78 + (945.71 \cdot 133.189/1725.223) / 150.0$$

$$= 0.588$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	150.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	75.0 mm.
Nominal Bolt Diameter	Bnd	27.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.1284 cm ²
Saddle Load Q0 (Weight)	Q0	26309.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1915.7 Kg
Maximum Transverse Force	Ft	2613.5 Kg



Maximum Longitudinal Force F1 5227.1 Kg
 Saddle Bolted to Steel Foundation No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 $= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 5227.1 / (3.128 \cdot 2)$
 $= 81.9 \text{ N./mm}^2$. Must be less than 100.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 2613.5 / (3.128 \cdot 2)$
 $= 41.0 \text{ N./mm}^2$. Must be less than 100.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 5227.1 / (100.0 \cdot 2.0)$
 $= 2.5631 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 1400.0 \cdot 2613.55 + 0.0$
 $= 3659.04 \text{ Kg-m.}$

Eccentricity (e):
 $= R_{mom} / QO$
 $= 3659.04 / 26309.04$
 $= 139.08 \text{ mm.}$ < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0.0, 2.5631, 0.0]$
 $= 2.5631 \text{ cm}^2$

MDMT Calculations for: FIXED SADDLE, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 18.0, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.986$, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (250.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(160.0 + 1.56 \cdot \sqrt{517.5 \cdot 74.0}, 2 \cdot 1650.0)$
 $= 465.2781 \text{ mm.}$

Input and Calculated Values:

Vessel Mean Radius	Rm	517.50 mm.
Shell Thickness used in this Case	t	77.000 mm.



Stiffened Vessel Length per 4.15.6 L 6840.00 mm.
 Distance from Saddle to Vessel tangent a1 or a 1650.00 mm.
 Saddle Width b1 or b 160.00 mm.
 Saddle Bearing Angle delta or theta 120.00 degrees
 Shell Allowable Stress used in Calculation 138.00 N./mm²
 Head Allowable Stress used in Calculation 138.00 N./mm²
 Circumferential Efficiency in Plane of Saddle 1.00
 Circumferential Efficiency at Mid-Span 1.00
 Distance from Saddle Base to Centerline B 1400.00 mm.
 Coefficient of Friction mu 0.10

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 40869.07 Kgf
 Pressure used in Analysis P 191.047 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	66.29	138.00
Long. Stress at Bottom of Midspan	67.32	138.00
Long. Stress at Top of Saddles	89.66	138.00
Long. Stress at Bottom of Saddles	54.14	138.00
Tangential Shear in Shell	6.34	110.40
Circ. Stress at Horn of Saddle	8.71	172.50
Circ. Compressive Stress in Shell	0.88	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (1368.5/2 + 0) \cdot 1400.0/991.5991$
 $= 2898.3 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(247.64, 0) \cdot 1400.0/3820.0005$
 $= 90.8 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(5227.1, 0) \cdot 1400.0/3820.0005$
 $= 1915.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (5227/2 + 0) \cdot 1400.0/991.5991$
 $= 11069.9 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 29799 + \max(91, 2898, 1916, 11070)$
 $= 40869.1 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (1.109 \cdot 1.2)^2)$
 $= 2428.479 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 41207.30 Kgf
 Transverse Shear Load 2613.55 Kgf
 Longitudinal Shear Load 5227.10 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(955.0/2 + 77.0 + 18.0)\sin(120.0/2), 2 \cdot 517.5)$$

$$= 991.599 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -40869 \cdot 1650.0 [1 - (1 - 1650.0/6840.0 + (517.5^2 - 0.0^2)/(2 \cdot 1650.0 \cdot 6840.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6840.0))]$$

$$= -15467.2 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 40869 \cdot 6840/4((1 + 2(517.5^2 - 0.0^2)/(6840^2))/(1 + (4 \cdot 0)/(3 \cdot 6840)) - 4 \cdot 1650/6840)$$

$$= 3252.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 191.047 \cdot 517.5/(2 \cdot 74.0) - 3252.3/(\pi \cdot 517.5^2 \cdot 74.0)$$

$$= 66.29 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 191.047 \cdot 517.5/(2 \cdot 74.0) + 3252.3/(\pi \cdot 517.5^2 \cdot 74.0)$$

$$= 67.32 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 191.047 \cdot 517.5/(2 \cdot 74.0) - 15467.2/(0.1066 \cdot \pi \cdot 517.5^2 \cdot 74.0)$$

$$= 89.66 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 191.047 \cdot 517.5/(2 \cdot 74.0) + 15467.2/(0.1923 \cdot \pi \cdot 517.5^2 \cdot 74.0)$$

$$= 54.14 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 40869(6840.0 - 2 \cdot 1650.0)/(6840.0 + (4 \cdot 0.0/3))$$

$$= 21151.5 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 21151.54/(517.5 \cdot 74.0)$$

$$= 6.34 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(517.5 \cdot 74.0)}$$

$$= 152.639 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 40869 \cdot 0.1/(74.0(160.0 + 152.64 + 152.64))$$

$$= -0.88 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q/(4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q/(2 \cdot t^2)$$

$$= -40869/(4 \cdot 74.0(160.0 + 152.639 + 152.639)) - 3 \cdot 0.0529 \cdot 40869/(2 \cdot 74.0^2)$$

$$= -8.71 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$
 $= \min(160.0 + 1.56 \cdot \sqrt{ 517.5 \cdot 74.0 }, 2 \cdot 1650.0)$
 $= 465.28 \text{ mm.}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1090.0000	mm.
Baseplate Thickness	Bpthk	32.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	18.0000	mm.
Web Thickness	Webtk	18.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		200.0	°C
Saddle Yield Stress	Sy	225.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{ R \cdot t } + \text{WearPlateWidth}) t^3) / 12$
 $= ((1.56 \cdot \sqrt{ 480.5 \cdot 74.0 } + 250.0) 74.0^3) / 12$
 $= 1837.562 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 1837.562 + 402.68 (223.839 - 37.0)^2$
 $= 142408.219 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	544.2	74.0	37.0	402.7	1489916.6	0.142E+06
Wearplate	250.0	18.0	83.0	45.0	373500.0	0.894E+04
Web	18.0	795.5	489.8	143.2	7012729.5	0.177E+06
BasePlate	200.0	32.0	903.5	64.0	5782399.5	0.296E+06
Totals	...	...	...	654.9	14658545.0	0.624E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 14658545 / 654.87$
 $= 223.839 \text{ mm.}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120.0 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 40869.07$
 $= 8317.7432 \text{ Kgf}$

Tension Stress, St = (Fh/As) = 3.2345 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 134.9988 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$
 $= 1400.0 - 480.5 \cdot \sin(120.0 / 2) / 1.0472$
 $= 1002.630 \text{ mm.}$

Bending Moment, M = Fh * d = 8339.7803 Kg-m.

Bending Stress, $S_b = (M \cdot C_1 / I) = 2.9347 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 149.9987 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(40869 + 338)200.0 / (2 \cdot 1090.0 \cdot 150.0))}$
 $= 27.230 \text{ mm.}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120.0/2) (477.5 + 77.0 + 18.0)$
 $= 991.599 \text{ mm.}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 991.5991 / (3 - 1)$
 $= 495.800 \text{ mm.}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 495.7996 \cdot 200.0 / 2$
 $= 495.800 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 41207.301 / (1090.0 \cdot 200.0)$
 $= 18.902 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 495.8 \cdot 18.9$
 $= 9371.817 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 25.56 + 44.622$
 $= 70.182 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 9371.8 / 70.182$
 $= 13.096 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	18.0	142.0	89.0	25.6	227484.0	0.109E+04
Web	247.9	18.0	9.0	44.6	40159.8	391.
Totals	...	...	...	70.2	267643.8	0.148E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 160.0 - 18.0$
 $= 142.000 \text{ mm.}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 267643.75 / 70.182$
 $= 38.136 \text{ mm.}$

Distance to Centroid [C1]:
 $= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$
 $= \max(38.136, 160.0 - 38.136)$
 $= 121.864 \text{ mm.}$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1481.6/70.182}$$

$$= 45.947 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1400.0 - \cos(120.0/2) (477.5 + 77.0 + 18.0) - 32.0$$

$$= 1081.750 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19258\text{E}+09/225.0)}$$

$$= 129.985$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1081.75/45.947$$

$$= 23.544$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 5227.1 / (2 \cdot 1090.0) \cdot 495.8 \cdot 1081.75/2$$

$$= 643.007 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ (23.5436 < 129.9855) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (23.54)^2 / (2 \cdot 129.99^2)) 225 /$$

$$(5/3 + 3 \cdot (23.54) / (8 \cdot 129.99) - (23.54)^3 / (8 \cdot 129.99^3))$$

$$= 127.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 13.1/127.64 + (643.01 \cdot 121.864/14816142) / 150.0$$

$$= 0.448$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	18.0	142.0	89.0	25.6	227484.0	0.142E+04
Web	495.8	18.0	9.0	89.2	80319.5	307.
Totals	...	...	...	114.8	307803.5	0.173E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 307803.5/114.804$$

$$= 26.811 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{abs}(\text{Saddle Width} - \text{ytot}))$$

$$= \max(26.811, \text{abs}(160.0 - 26.811))$$

$$= 133.189 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1400.0 - \cos(572.5/2) (32.0 + 0.0 + 0.0) - 0.0$$

$$= 795.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1725.2/114.804}$$

$$= 38.765 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 795.5/38.765$$

$$= 20.521$$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 5227.1 / (2 \cdot 1090.0)$
 $= 2.398 \text{ Kg/mm.}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 2.398 \cdot 495.8 \cdot 795.5$
 $= 945.713 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 945.713 / 129532.297$
 $= 71.598 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ (20.5209 < 129.9855) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (20.52)^2 / (2 \cdot 129.99^2)) 225 /$
 $(5/3 + 3 \cdot (20.52) / (8 \cdot 129.99) - (20.52)^3 / (8 \cdot 129.99^3))$
 $= 128.8 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 15.65/128.78 + (945.71 \cdot 133.189 / 1725.223) / 150.0$
 $= 0.599$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	150.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	75.0 mm.
Nominal Bolt Diameter	Bnd	27.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.1284 cm ²
Saddle Load Q0 (Weight)	Q0	30137.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1915.7 Kg
Maximum Transverse Force	Ft	2613.5 Kg
Maximum Longitudinal Force	Fl	5227.1 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 5227.1 / (3.128 \cdot 2)$
 $= 81.9 \text{ N./mm}^2$. Must be less than 100.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 2613.5 / (3.128 \cdot 2)$
 $= 41.0 \text{ N./mm}^2$. Must be less than 100.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 5227.1 / (100.0 \cdot 2.0)$
 $= 2.5631 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$



$$= 1400.0 \cdot 2613.55 + 0.0$$
$$= 3659.04 \text{ Kg-m.}$$

Eccentricity (e):
= Rmom / QO
= 3659.04/30137.4
= 121.41 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 2.5631, 0.0]
= 2.5631 cm²

MDMT Calculations for: SLIDING SADDLE, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 18.0, c = 3.0 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.986, Temp. Reduction = 1 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (250.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$$

$$= \min(160.0 + 1.56 \cdot \sqrt{ 517.5 \cdot 74.0 }, 2 \cdot 1370.0)$$

$$= 465.2781 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	517.50	mm.
Shell Thickness used in this Case	t	77.000	mm.
Stiffened Vessel Length per 4.15.6	L	6840.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1370.00	mm.
Saddle Width	b1 or b	160.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1400.00	mm.
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		26128.26	Kgf
Pressure used in Analysis	P	249.614	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	85.81	234.00
Long. Stress at Bottom of Midspan	88.77	234.00
Long. Stress at Top of Saddles	97.12	234.00
Long. Stress at Bottom of Saddles	81.83	234.00
Tangential Shear in Shell	4.70	138.58
Circ. Stress at Horn of Saddle	5.57	350.99
Circ. Compressive Stress in Shell	0.57	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(451.6/2 + 0) \cdot 1400.0/991.5991$$

$$= 956.4 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl, wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(81.72, 0) \cdot 1400.0/3820.0005$$

$$= 30.0 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 25172 + \max(30, 956, 0, 0)$$

$$= 26128.3 \text{ Kgf}$$

Longitudinal Wind Force [FI]:

$$\begin{aligned}
 &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2)) \\
 &= 1.1 \cdot 1587.169(0.0 + \pi/4(1.109 \cdot 1.2)^2) \\
 &= 2428.479 \text{ N}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	26466.49	Kgf
Transverse Shear Load	225.81	Kgf
Longitudinal Shear Load	81.72	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(955.0/2 + 77.0 + 18.0)\sin(120.0/2), 2 \cdot 517.5) \\
 &= 991.599 \text{ mm.}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\
 &= -26128 \cdot 1370.0 [1 - (1 - 1370.0/6840.0 + (517.5^2 - 0.0^2)/(2 \cdot 1370.0 \cdot 6840.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6840.0))] \\
 &= -6658.2 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L) \\
 &= 26128 \cdot 6840/4((1 + 2(517.5^2 - 0^2)/(6840^2))/(1 + (4 \cdot 0)/(3 \cdot 6840)) - 4 \cdot 1370/6840) \\
 &= 9395.3 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\
 &= 249.614 \cdot 517.5/(2 \cdot 74.0) - 9395.3/(\pi \cdot 517.5^2 \cdot 74.0) \\
 &= 85.81 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\
 &= 249.614 \cdot 517.5/(2 \cdot 74.0) + 9395.3/(\pi \cdot 517.5^2 \cdot 74.0) \\
 &= 88.77 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\
 &= 249.614 \cdot 517.5/(2 \cdot 74.0) - 6658.2/(0.1066 \cdot \pi \cdot 517.5^2 \cdot 74.0) \\
 &= 97.12 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 249.614 \cdot 517.5/(2 \cdot 74.0) + 6658.2/(0.1923 \cdot \pi \cdot 517.5^2 \cdot 74.0) \\
 &= 81.83 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\
 &= 26128(6840.0 - 2 \cdot 1370.0)/(6840.0 + (4 \cdot 0.0/3)) \\
 &= 15661.7 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (R_m \cdot t) \\
 &= 1.1707 \cdot 15661.68/(517.5 \cdot 74.0) \\
 &= 4.70 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(R_m \cdot t)} \\
 &= 0.78 \cdot \sqrt{(517.5 \cdot 74.0)} \\
 &= 152.639 \text{ mm.}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 $= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$
 $= -0.7603 \cdot 26128 \cdot 0.1 / (74.0(160.0 + 152.64 + 152.64))$
 $= -0.57 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:
 $= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$
 $= -26128 / (4 \cdot 74.0(160.0 + 152.639 + 152.639)) -$
 $3 \cdot 0.0529 \cdot 26128 / (2 \cdot 74.0^2)$
 $= -5.57 \text{ N./mm}^2$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(160.0 + 1.56 \cdot \sqrt{517.5 \cdot 74.0}, 2 \cdot 1370.0)$
 $= 465.28 \text{ mm.}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1090.0000	mm.
Baseplate Thickness	Bpthk	32.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	18.0000	mm.
Web Thickness	Webtk	18.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3) / 12$
 $= ((1.56 \cdot \sqrt{480.5 \cdot 74.0} + 250.0)74.0^3) / 12$
 $= 1837.562 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 $= 1837.562 + 402.68(223.839 - 37.0)^2$
 $= 142408.219 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	544.2	74.0	37.0	402.7	1489916.6	0.142E+06
Wearplate	250.0	18.0	83.0	45.0	373500.0	0.894E+04
Web	18.0	795.5	489.8	143.2	7012729.5	0.177E+06
BasePlate	200.0	32.0	903.5	64.0	5782399.5	0.296E+06
Totals	...	...	...	654.9	14658545.0	0.624E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 14658545 / 654.87$
 $= 223.839 \text{ mm.}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120.0 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K_1 \cdot Q$
 $= 0.204 \cdot 26128.258$
 $= 5317.6670 \text{ Kg}$

Tension Stress, $St = (Fh / A_s) = 2.0679 \text{ N./mm}^2$

Allowed Stress, $S_a = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1400.0 - 480.5 \cdot \sin(120.0/2) / 1.0472$
 $= 1002.630 \text{ mm.}$

Bending Moment, $M = F_h \cdot d = 5331.7559 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 1.8762 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt{3(26128 + 338)200.0 / (2 \cdot 1090.0 \cdot 173.33)}$
 $= 20.301 \text{ mm.}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120.0/2) (477.5 + 77.0 + 18.0)$
 $= 991.599 \text{ mm.}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 991.5991 / (3 - 1)$
 $= 495.800 \text{ mm.}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 495.7996 \cdot 200.0 / 2$
 $= 495.800 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 26466.484 / (1090.0 \cdot 200.0)$
 $= 12.141 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 495.8 \cdot 12.14$
 $= 6019.299 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 25.56 + 44.622$
 $= 70.182 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 6019.3 / 70.182$
 $= 8.411 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	18.0	142.0	89.0	25.6	227484.0	0.109E+04
Web	247.9	18.0	9.0	44.6	40159.8	391.
Totals	...	...	...	70.2	267643.8	0.148E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 160.0 - 18.0$
 $= 142.000 \text{ mm.}$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 267643.75/70.182$$

$$= 38.136 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(38.136, 160.0 - 38.136)$$

$$= 121.864 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 1481.6/70.182 }$$

$$= 45.947 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1400.0 - \cos(120.0/2) (477.5 + 77.0 + 18.0) - 32.0$$

$$= 1081.750 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19258\text{E}+09/260.0)}$$

$$= 120.921$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1081.75/45.947$$

$$= 23.544$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 81.7 / (2 \cdot 1090.0) \cdot 495.8 \cdot 1081.75/2$$

$$= 10.053 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($23.5436 < 120.9209$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (23.54)^2 / (2 \cdot 120.92^2)) 260 / (5/3 + 3 \cdot (23.54) / (8 \cdot 120.92) - (23.54)^3 / (8 \cdot 120.92^3))$$

$$= 146.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 8.41/146.7 + (10.05 \cdot 121.864/14816142) / 173.33$$

$$= 0.062$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	18.0	142.0	89.0	25.6	227484.0	0.142E+04
Web	495.8	18.0	9.0	89.2	80319.5	307.
Totals	...	...	...	114.8	307803.5	0.173E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 307803.5/114.804$$

$$= 26.811 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs(Saddle Width} - y_{tot}))$$

$$= \max(26.811, \text{abs(160.0} - 26.811))$$

$$= 133.189 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1400.0 - \cos(572.5/2) (32.0 + 0.0 + 0.0) - 0.0$$

$$= 795.500 \text{ mm.}$$

Radius of Gyration [r]:



$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1725.2/114.804}$$

$$= 38.765 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 795.5/38.765$$

$$= 20.521$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 81.721/(2 \cdot 1090.0)$$

$$= 0.037 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.037 \cdot 495.8 \cdot 795.5$$

$$= 14.785 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 14.785/129532.297$$

$$= 1.119 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($20.5209 < 120.9209$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (20.52)^2 / (2 \cdot 120.92^2)) 260 /$$

$$(5/3 + 3 \cdot (20.52) / (8 \cdot 120.92) - (20.52)^3 / (8 \cdot 120.92^3))$$

$$= 148.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$$

$$= 10.05/148.15 + (14.79 \cdot 133.189/1725.223)/173.33$$

$$= 0.074$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	75.0 mm.
Nominal Bolt Diameter	Bnd	27.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	3.1284 cm ²
Saddle Load Q0 (Weight)	Q0	25510.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	30.0 Kg
Maximum Transverse Force	Ft	225.8 Kg
Maximum Longitudinal Force	Fl	81.7 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 81.7/(3.128 \cdot 2)$$

$$= 1.3 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 225.8/(3.128 \cdot 2)$$

$$= 3.5 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 81.72 / (68.4 \cdot 2.0)$$

$$= 0.0586 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1400.0 \cdot 225.81 + 0.0$$

$$= 316.14 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 316.14 / 25510.06$$

$$= 12.39 \text{ mm.} < B_{plen}/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.0586, 0.0]$$

$$= 0.0586 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (250.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(160.0 + 1.56 \cdot \sqrt{517.5 \cdot 74.0}, 2 \cdot 1650.0)$$

$$= 465.2781 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	517.50	mm.
Shell Thickness used in this Case	t	77.000	mm.
Stiffened Vessel Length per 4.15.6	L	6840.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1650.00	mm.
Saddle Width	b1 or b	160.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1400.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		30007.97	Kgf
Pressure used in Analysis	P	249.614	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		86.91	234.00
Long. Stress at Bottom of Midspan		87.66	234.00
Long. Stress at Top of Saddles		104.06	234.00
Long. Stress at Bottom of Saddles		77.99	234.00
Tangential Shear in Shell		4.66	138.58
Circ. Stress at Horn of Saddle		6.40	350.99
Circ. Compressive Stress in Shell		0.65	234.00



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (451.6/2 + 0) \cdot 1400.0/991.5991$
 $= 956.4 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(81.72, 0) \cdot 1400.0/3820.0005$
 $= 30.0 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 29052 + \max(30, 956, 0, 0)$
 $= 30008.0 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (1.109 \cdot 1.2)^2)$
 $= 2428.479 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	30346.20	Kgf
Transverse Shear Load	225.81	Kgf
Longitudinal Shear Load	81.72	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(955.0/2 + 77.0 + 18.0) \sin(120.0/2), 2 \cdot 517.5)$
 $= 991.599 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -30008 \cdot 1650.0 [1 - (1 - 1650.0/6840.0 + (517.5^2 - 0.0^2) / (2 \cdot 1650.0 \cdot 6840.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 6840.0))]$
 $= -11356.7 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 30008 \cdot 6840/4 ((1 + 2(517.5^2 - 0^2)/(6840^2)) / (1 + (4 \cdot 0)/(3 \cdot 6840)) - 4 \cdot 1650/6840)$
 $= 2388.0 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 249.614 \cdot 517.5/(2 \cdot 74.0) - 2388.0/(\pi \cdot 517.5^2 \cdot 74.0)$
 $= 86.91 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 249.614 \cdot 517.5/(2 \cdot 74.0) + 2388.0/(\pi \cdot 517.5^2 \cdot 74.0)$
 $= 87.66 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 $= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$
 $= 249.614 \cdot 517.5/(2 \cdot 74.0) - 11356.7/(0.1066 \cdot \pi \cdot 517.5^2 \cdot 74.0)$
 $= 104.06 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 249.614 \cdot 517.5 / (2 \cdot 74.0) + 11356.7 / (0.1923 \cdot \pi \cdot 517.5^2 \cdot 74.0)$$

$$= 77.99 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 30008(6840.0 - 2 \cdot 1650.0) / (6840.0 + (4 \cdot 0.0 / 3))$$

$$= 15530.4 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 15530.44 / (517.5 \cdot 74.0)$$

$$= 4.66 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(517.5 \cdot 74.0)}$$

$$= 152.639 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 30008 \cdot 0.1 / (74.0(160.0 + 152.64 + 152.64))$$

$$= -0.65 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -30008 / (4 \cdot 74.0(160.0 + 152.639 + 152.639)) - 3 \cdot 0.0529 \cdot 30008 / (2 \cdot 74.0^2)$$

$$= -6.40 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(160.0 + 1.56 \cdot \sqrt{(517.5 \cdot 74.0)}, 2 \cdot 1650.0)$$

$$= 465.28 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1090.0000	mm.
Baseplate Thickness	Bpthk	32.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	18.0000	mm.
Web Thickness	Webtk	18.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(480.5 \cdot 74.0)} + 250.0)74.0^3) / 12$$

$$= 1837.562 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 1837.562 + 402.68(223.839 - 37.0)^2$$

$$= 142408.219 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	544.2	74.0	37.0	402.7	1489916.6	0.142E+06
Wearplate	250.0	18.0	83.0	45.0	373500.0	0.894E+04
Web	18.0	795.5	489.8	143.2	7012729.5	0.177E+06
BasePlate	200.0	32.0	903.5	64.0	5782399.5	0.296E+06
Totals	...	...	...	654.9	14658545.0	0.624E+06

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 14658545/654.87 \\
 &= 223.839 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120.0/2 \\
 &= 120.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.204 \cdot 30007.971 \\
 &= 6107.2729 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 2.3749 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1400.0 - 480.5 \cdot \sin(120.0/2) / 1.0472 \\
 &= 1002.630 \text{ mm.}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 6123.4541 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 2.1548 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}) } \\
 &= \sqrt{ 3(30008 + 338)200.0 / (2 \cdot 1090.0 \cdot 173.33) } \\
 &= 21.738 \text{ mm.}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0/2) (477.5 + 77.0 + 18.0) \\
 &= 991.599 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 991.5991 / (3 - 1) \\
 &= 495.800 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot \text{Bpwid} / 2 \\
 &= 495.7996 \cdot 200.0/2 \\
 &= 495.800 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 30346.197 / (1090.0 \cdot 200.0) \\
 &= 13.920 \text{ Kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 495.8 \cdot 13.92 \\
 &= 6901.667 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of the Rib and Web [Ar]:} \\
 &= \text{Rib Area} + \text{Web Area} \\
 &= 25.56 + 44.622 \\
 &= 70.182 \text{ cm}^2
 \end{aligned}$$

$$\text{Compressive Stress [Sc]:}$$

= P/Ar
 = 6901.7/70.182
 = 9.644 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	18.0	142.0	89.0	25.6	227484.0	0.109E+04
Web	247.9	18.0	9.0	44.6	40159.8	391.
Totals	...	...	...	70.2	267643.8	0.148E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 160.0 - 18.0
 = 142.000 mm.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 267643.75/70.182
 = 38.136 mm.

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(38.136, 160.0 - 38.136)
 = 121.864 mm.

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{(1481.6/70.182)}$
 = 45.947 mm.

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1400.0 - cos(120.0/2)(477.5 + 77.0 + 18.0) - 32.0
 = 1081.750 mm.

Intermediate Term [Cc]:
 = $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 0.19258\text{E}+09/260.0)}$
 = 120.921

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1081.75/45.947
 = 23.544

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 81.7 / (2 • 1090.0) • 495.8 • 1081.75/2
 = 10.053 Kg-m.

Compressive Allowable, KL/r < Cc (23.5436 < 120.9209) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (23.54)^2 / (2 \cdot 120.92^2)) 260 /$
 $(5/3 + 3 \cdot (23.54) / (8 \cdot 120.92) - (23.54)^3 / (8 \cdot 120.92^3))$
 = 146.7 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba
 = 9.64/146.7 + (10.05 • 121.864/14816142) / 173.33
 = 0.070

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



Rib	18.0	142.0	89.0	25.6	227484.0	0.142E+04
Web	495.8	18.0	9.0	89.2	80319.5	307.
Totals	...	...	...	114.8	307803.5	0.173E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 307803.5 / 114.804$$

$$= 26.811 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(26.811, \text{abs}(160.0 - 26.811))$$

$$= 133.189 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1400.0 - \cos(572.5/2) (32.0 + 0.0 + 0.0) - 0.0$$

$$= 795.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1725.2 / 114.804) }$$

$$= 38.765 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 795.5 / 38.765$$

$$= 20.521$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 81.721 / (2 \cdot 1090.0)$$

$$= 0.037 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.037 \cdot 495.8 \cdot 795.5$$

$$= 14.785 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 14.785 / 129532.297$$

$$= 1.119 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (20.5209 < 120.9209) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (20.52)^2 / (2 \cdot 120.92^2)) 260 /$$

$$(5/3 + 3 \cdot (20.52) / (8 \cdot 120.92) - (20.52^3) / (8 \cdot 120.92^3))$$

$$= 148.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 11.53 / 148.15 + (14.79 \cdot 133.189 / 1725.223) / 173.33$$

$$= 0.084$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	75.0 mm.
Nominal Bolt Diameter	Bnd	27.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.1284 cm ²
Saddle Load Q0 (Weight)	Q0	29389.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	30.0 Kg
Maximum Transverse Force	Ft	225.8 Kg



Maximum Longitudinal Force F1 81.7 Kgf
Saddle Bolted to Steel Foundation No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$\begin{aligned} &= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 81.7 / (3.128 \cdot 2) \\ &= 1.3 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2. \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$\begin{aligned} &= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 225.8 / (3.128 \cdot 2) \\ &= 3.5 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2. \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $81.72 / (68.4 \cdot 2.0)$
= 0.0586 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1400.0 \cdot 225.81 + 0.0$
= 316.14 Kg-m.

Eccentricity (e):
= R_{mom} / QO
= $316.14 / 29389.78$
= 10.76 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.0586, 0.0]
= 0.0586 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: TS OUT NOZ. T2

From: 30

Pressure for Reinforcement Calculations	P	147.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	955.00	mm.
Shell Finished (Minimum) Thickness	t	60.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1158.17	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	40.4900	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	145.0000	mm.
Height of Beveled Transition	L'	37.2448	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		105.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		60.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: TS OUT NOZ. T2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	16.000	in.
Actual Thickness Used in Calculation	1.594	in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 147.0 \cdot 480.5 / (138 \cdot 1.0 - 0.6 \cdot 147.0) \\
 &= 54.6828 \text{ mm.}
 \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 147.0 \cdot 206.2 / (138 \cdot 1.0 - 0.6 \cdot 147.0) \\
 &= 23.4664 \text{ mm.}
 \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$\begin{aligned}
 &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\
 &= 64.51 / 0.5773 \\
 &= 111.7346 \text{ mm.}
 \end{aligned}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	824.8000	mm.
Parallel to Vessel Wall, opening length	d	412.4000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	142.5000	mm.

Weld Strength Reduction Factor [fr1]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 138.0 / 138.0) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr2]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 138.0 / 138.0) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr3]:

$$\begin{aligned}
 &= \min(fr2, fr4) \\
 &= \min(1.0, 1.0) \\
 &= 1.000
 \end{aligned}$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	225.512	NA	NA
Area in Shell	A1	9.556	NA	NA
Area in Nozzle Wall	A2	39.967	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	183.854	NA	NA
TOTAL AREA AVAILABLE	Atot	234.377	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (412.4 \cdot 54.6828 \cdot 1.0 + 2 \cdot 102.0 \cdot 54.6828 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 225.512 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 412.4(1.0 \cdot 57.0 - 1.0 \cdot 54.683) - 2 \cdot 102.0$$

$$(1.0 \cdot 57.0 - 1.0 \cdot 54.6828) \cdot (1 - 1.0)$$

$$= 9.556 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np})(t_n - t_{rn})fr_2$$

$$= (2 \cdot 142.5)(37.49 - 23.47)1.0$$

$$= 39.967 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_n/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(142.5, 300.0, 145.0)) \cdot (105.0 - 40.5) \cdot 1.0$$

$$= 183.854 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 26.4664 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 57.6828 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 57.6828 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(57.6828, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(26.4664, 11.3312)$$

$$= 26.4664 \text{ mm.}$$

Available Nozzle Neck Thickness = 40.4900 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 68.6,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 276.4 N./mm ²	Passed
Occasional	: 37.1,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 21.9,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 40.49, tr = 23.466, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.626, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	11 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 60.0, tr = 54.683, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.959, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	19 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 19 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 147.00/255.10 = 0.576

Weld Size Calculations, Description: TS OUT NOZ. T2

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (225.5119 - 9.5561 + 2 \cdot 102.0 \cdot 1.0 \cdot (1.0 \cdot 57.0 - 54.6828)) \cdot 138)$
 $= 310535.44 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (223.8208 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 316358.19 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (223.8208 + 0.0 + 1. + (116.28)) \cdot 138$
 $= 479982.44 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (223.8208 + 0.0 + 1. + 0.0 + (116.28)) \cdot 138$
 $= 479982.44 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 616.4 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 66761. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 257.2) \cdot (105.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 811823. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 616.4 \cdot (60.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 574687. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (66761 + 811823) = 878584 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (66761 + 0 + 574687 + 0) = 641448 \text{ Kgf}$

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (66761 + 574687 + 0) = 641448 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 878584 Kgf, must exceed W = 310535 Kgf or W1 = 316358 Kgf
 Path 2-2 = 641448 Kgf, must exceed W = 310535 Kgf or W2 = 479982 Kgf
 Path 3-3 = 641448 Kgf, must exceed W = 310535 Kgf or W3 = 479982 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 149 bars

The Drop for this Nozzle is : 112.7823 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 472.7823 mm.

Input Echo, WRC297 Item 1, Description: TS OUT NOZ. T2:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	955.000	mm.
Shell Thickness	60.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	406.400	mm.
Nozzle Thickness	105.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	147.00	bars
Radial Load	P	-13053.20	Kgf
Circumferential Shear	Vc	9789.91	Kgf
Longitudinal Shear	Vl	13053.20	Kgf
Circumferential Moment	Mc	13570.10	Kg-m.
Longitudinal Moment	Ml	13570.10	Kg-m.
Torsional Moment	Mt	15657.81	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (17.917) is < 20
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Dv (.573) is > 0.5
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (5.870) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 = NozzleOD + 2 • 1.65 • √(Rmean(t - ca))



$$= 616.4 + 2 \cdot 1.65 \cdot \sqrt{(509.0(60.0 - 3.0))}$$

$$= 1178.496 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1018.000 mm.
 Nozzle Outside Diameter (d) = 616.400 mm.
 Vessel Thickness used (T) = 57.000 mm.
 Nozzle Thickness used (t) = 102.000 mm.
 T / t = 0.559
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.559$$

Nr/P = 0.066
 Mr/P = 0.118
 M0/P = 0.037
 N0/P = 0.091
 MrD/Mc = 0.269
 NrDL/Mc = 0.106
 M0d/Mc = 0.083
 N0DL/Mc = 0.077
 MrD/M1 = 0.102
 NrDL/M1 = 0.056
 M0D/M1 = 0.030
 N0DL/M1 = 0.109

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	116	131	116	131
Outplane Stress Summary	109	131	148	121

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-40	40	40	-40
Inplane Pressure Stress	58	58	58	58
Inplane Stress Summary	44	70	133	-3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stress Intensities	Top Outside	Top Inside	Bottom Outside	Bottom Inside
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Two * Max Shear Stress	110	132	149	125
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-33	33	33	-33
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58

Outplane Stress Summary	32	81	109	25
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-107	107	107	-107
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	116	131	116	131

Inplane Stress Summary	32	206	261	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	34	206	262	27
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	46	28	46

Hoop Stress Summary	24	42	39	57
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-9	9	9	-9
Axial Pressure Stress	14	14	14	14
Axial Stress Summary	4	10	37	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	25	42	40	57

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	46	28	46
Hoop Stress Summary	27	44	37	54

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-26	26	26	-26
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	14	14	14	14
Axial Stress Summary	-12	28	55	-10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (N./mm²):



CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	39	44	56	66

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	131	116	131	116	131	116	131
Circ. Pl (SUS)	-4	-4	10	10	-5	-5	9	9
Circ. Q (SUS)	-4	4	20	-20	-80	80	134	-134
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	-1	-1	5	5	-2	-2	8	8
Long. Q (SUS)	-13	13	67	-67	-25	25	41	-41
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	116.0	131.0	116.0	131.0	116.0	131.0	116.0	131.0
Pm+Pl (SUS)	112.0	127.0	126.0	141.0	111.1	126.1	125.1	140.1
Pm+Pl+Q (Total)	108.4	131.4	146.5	125.1	33.0	206.0	259.2	26.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+Pl (SUS)	141.01	207.00	Passed
Pm+Pl+Q (TOTAL)	259.24	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	28	46	28	46	28	46	28	46
Circ. Pl (SUS)	-4	-4	10	10	-2	-2	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-3	3	15	-15	-20	20	32	-32
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	28.0	46.0	28.0	46.0	28.0	46.0	28.0	46.0
Pm+Pl (SUS)	24.0	42.0	38.0	56.0	26.1	44.0	36.1	54.0
Pm+Pl+Q (Total)	24.2	42.1	39.2	56.1	39.1	44.1	53.5	65.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	138.00	Passed
Pm+Pl (SUS)	56.00	207.00	Passed
Pm+Pl+Q (TOTAL)	65.28	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: DRAIN NOZ. D1

From: 30

Pressure for Reinforcement Calculations	P	147.094	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	955.00	mm.
Shell Finished (Minimum) Thickness	t	60.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1258.18	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	60.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: DRAIN NOZ. D1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687	in.
Actual Thickness Used in Calculation	0.344	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147.09 \cdot 480.5 / (138 \cdot 1.0 - 0.6 \cdot 147.09)$
 $= 54.7200$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147.09 \cdot 24.42 / (138 \cdot 1.0 - 0.6 \cdot 147.09)$

= 2.7815 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	174.3250	mm.
Parallel to Vessel Wall	Rn+tn+t	87.1625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	14.3440	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: DRAIN NOZ. D1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.7815	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 57.7200	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 57.7200	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 6.4200	mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[6.42, max(57.72, 4.5)]
 = 6.4200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(5.7815, 6.42)
 = 6.4200 mm.

Available Nozzle Neck Thickness = 0.875 * 8.738 = 7.645 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 7.645, tr = 2.782, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.599, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Weld Size Calculations, Description: DRAIN NOZ. D1

Intermediate Calc. for nozzle/shell Welds	Tmin	5.7376	mm.
---	------	--------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.0163 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 0.9536 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 360.9536 mm.

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Input, Nozzle Desc: SH. IN NOZ. S1

From: 40

Pressure for Reinforcement Calculations	P	191.000	bars
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	955.00	mm.
Shell Finished (Minimum) Thickness	t	77.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2838.17	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	60.0000	mm.
Flange Material		SA-350 LF2 +IT	
Hub Height of Integral Nozzle	h	150.0000	mm.
Height of Beveled Transition	L'	63.7741	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		175.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		77.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: SH. IN NOZ. S1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	2.362	in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 191.0 \cdot 480.5 / (138 \cdot 1.0 - 0.6 \cdot 191.0) \\
 &= 72.5329 \text{ mm.}
 \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 191.0 \cdot 307.8 / (138 \cdot 1.0 - 0.6 \cdot 191.0) \\
 &= 46.4633 \text{ mm.}
 \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$\begin{aligned}
 &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\
 &= 115.0 / 0.5773 \\
 &= 199.1859 \text{ mm.}
 \end{aligned}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1231.2001 mm.
Parallel to Vessel Wall, opening length	d	615.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	185.0000 mm.

Weld Strength Reduction Factor [fr1]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 138.0 / 138.0) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr2]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 138.0 / 138.0) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr3]:

$$\begin{aligned}
 &= \min(fr2, fr4) \\
 &= \min(1.0, 1.0) \\
 &= 1.000
 \end{aligned}$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	446.513	NA	NA
Area in Shell	A1	9.031	NA	NA
Area in Nozzle Wall	A2	38.986	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	403.410	NA	NA
TOTAL AREA AVAILABLE	Atot	452.428	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (615.6 \cdot 72.5329 \cdot 1.0 + 2 \cdot 172.0 \cdot 72.5329 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 446.513 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$



$$= 615.6(1.0 \cdot 74.0 - 1.0 \cdot 72.533) - 2 \cdot 172.0$$

$$(1.0 \cdot 74.0 - 1.0 \cdot 72.5329) \cdot (1 - 1.0)$$

$$= 9.031 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np})(t_n - t_{rn})fr_2$$

$$= (2 \cdot 185.0)(57.0 - 46.46)1.0$$

$$= 38.986 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(185.0, 200.0, 150.0)) \cdot (175.0 - 60.0) \cdot 1.0$$

$$= 403.410 \text{ cm}^2 \text{ Includes Bevel Area of } 58.410 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 49.4633 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 75.5329 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 75.5329 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(75.5329, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(49.4633, 11.3312)$$

$$= 49.4633 \text{ mm.}$$

Available Nozzle Neck Thickness = 60.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 68.1,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 276.9 N./mm ²	Passed
Occasional	: 47.2,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 14.5,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 77.0, tr = 72.533, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.98, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	25 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 25 °C

Flange MDMT including Temperature reduction per UCS-66.1:



MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -60 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 191.00/255.10 = 0.749

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 96.9/0.5773$
 $= 167.8358 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 923.4000 mm.
 Parallel to Vessel Wall, opening length d 461.7000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 185.0000 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	297.690	297.690
Area Available in Shell	A1	4.516	4.516
Area Available in Nozzle Wall	A2	38.986	38.986
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	-0.000	-0.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	339.917	339.917
TOTAL AREA AVAILABLE	Atot	383.419	383.419

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: SH. IN NOZ. S1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (446.5125 - 9.0315 + 2 \cdot 172.0 \cdot 1.0 \cdot (1.0 \cdot 74.0 - 72.5329)) \cdot 138)$
 $= 622706.19 \text{ Kg}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (378.9028 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 534582.88 \text{ Kg}$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (378.9028 + 0.0 + 1. + (254.56)) \cdot 138$$

$$= 892788.75 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (378.9028 + 0.0 + 1. + 0.0 + (254.56)) \cdot 138$$

$$= 892788.75 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 959.5999 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 103932. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 393.8) \cdot (175.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 2096016. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 959.5999 \cdot (77.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 1161492. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (103932 + 2096016) = 2199948 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (103932 + 0 + 1161492 + 0) = 1265424 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (103932 + 1161492 + 0) = 1265424 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 2199948 Kgf, must exceed W = 622706 Kgf or W1 = 534582 Kgf
 Path 2-2 = 1265423 Kgf, must exceed W = 622706 Kgf or W2 = 892788 Kgf
 Path 3-3 = 1265423 Kgf, must exceed W = 622706 Kgf or W3 = 892788 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 192 bars

The Drop for this Nozzle is : 477.5000 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 754.5000 mm.

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: SH. IN NOZ. S1:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	955.000	mm.
Shell Thickness	77.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	609.600	mm.
Nozzle Thickness	175.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	191.00	bars
Radial Load	P	-19579.92	Kgf
Circumferential Shear	Vc	14684.89	Kgf
Longitudinal Shear	Vl	19579.92	Kgf
Circumferential Moment	Mc	30532.79	Kg-m.
Longitudinal Moment	Ml	30532.79	Kg-m.
Torsional Moment	Mt	35230.17	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (14.403) is < 20
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Dv (.865) is > 0.5
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (5.483) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 959.6 + 2 \cdot 1.65 \cdot \sqrt{(517.5(77.0 - 3.0))}$
 $= 1605.380 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1035.000 mm.
Nozzle Outside Diameter	(d) =	959.600 mm.
Vessel Thickness used	(T) =	74.000 mm.
Nozzle Thickness used	(t) =	172.000 mm.
T / t	=	0.430
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.467$$

Nr/P	=	0.059
Mr/P	=	0.090
M0/P	=	0.031
N0/P	=	0.079
MrD/Mc	=	0.260
NrDL/Mc	=	0.098
M0d/Mc	=	0.078
N0DL/Mc	=	0.086
MrD/Ml	=	0.086
NrDL/Ml	=	0.048
M0D/Ml	=	0.024
N0DL/Ml	=	0.086

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom



longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	115	134	115	134
Outplane Stress Summary	111	133	137	127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-29	29	29	-29
Inplane Pressure Stress	57	57	57	57
Inplane Stress Summary	46	67	110	14

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	111	134	137	127

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-26	26	26	-26
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	57	57	57	57
Outplane Stress Summary	35	75	98	31

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-88	88	88	-88
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	115	134	115	134
Inplane Stress Summary	41	200	230	33

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	42	200	230	38

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	32	54	32	54
Hoop Stress Summary	29	52	39	62

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	10	13	26	14

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	30	52	39	62

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	32	54	32	54
Hoop Stress Summary	29	52	39	62

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-13	13	13	-13
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	1	23	36	4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	2	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	29	52	40	62

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	115	134	115	134	115	134	115	134
Circ. Pl (SUS)	-2	-2	6	6	-3	-3	7	7
Circ. Q (SUS)	-2	2	14	-14	-69	69	107	-107
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	0	0	4	4	-2	-2	6	6
Long. Q (SUS)	-10	10	48	-48	-20	20	32	-32
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	115.0	134.0	115.0	134.0	115.0	134.0	115.0	134.0
Pm+Pl (SUS)	113.0	132.0	121.0	140.0	112.0	131.0	122.0	141.0
Pm+Pl+Q (Total)	111.2	134.2	135.2	126.0	43.5	200.0	229.1	36.8



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	134.00	138.00	Passed
Pm+Pl (SUS)	141.01	207.00	Passed
Pm+Pl+Q (TOTAL)	229.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	54	32	54	32	54	32	54
Circ. Pl (SUS)	-2	-2	6	6	-2	-2	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-4	-4	4	4	-4	-4	4	4
Long. Q (SUS)	-1	1	5	-5	-11	11	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	32.0	54.0	32.0	54.0	32.0	54.0	32.0	54.0
Pm+Pl (SUS)	30.0	52.0	38.0	60.0	30.0	52.0	38.0	60.0
Pm+Pl+Q (Total)	30.1	52.0	38.1	60.0	30.0	52.0	38.3	60.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	54.00	138.00	Passed
Pm+Pl (SUS)	60.00	207.00	Passed
Pm+Pl+Q (TOTAL)	60.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: SH. OUT NOZ. S2

From: 40

Pressure for Reinforcement Calculations	P	191.094	bars
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	955.00	mm.
Shell Finished (Minimum) Thickness	t	77.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		8072.17	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	59.5400	mm.
Flange Material		SA-350 LF2 +IT	
Hub Height of Integral Nozzle	h	150.0000	mm.
Height of Beveled Transition	L'	66.6608	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		175.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		77.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: SH. OUT NOZ. S2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	2.344	in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 191.09 \cdot 480.5 / (138 \cdot 1.0 - 0.6 \cdot 191.09) \\
 &= 72.5717 \text{ mm.}
 \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 191.09 \cdot 307.8 / (138 \cdot 1.0 - 0.6 \cdot 191.09) \\
 &= 46.4882 \text{ mm.}
 \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$\begin{aligned}
 &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\
 &= 115.46 / 0.5773 \\
 &= 199.9826 \text{ mm.}
 \end{aligned}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1231.2001 mm.
Parallel to Vessel Wall, opening length	d	615.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	185.0000 mm.

Weld Strength Reduction Factor [fr1]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 138.0 / 138.0) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr2]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 138.0 / 138.0) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr3]:

$$\begin{aligned}
 &= \min(fr2, fr4) \\
 &= \min(1.0, 1.0) \\
 &= 1.000
 \end{aligned}$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	446.751	NA	NA
Area in Shell	A1	8.793	NA	NA
Area in Nozzle Wall	A2	37.192	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	405.984	NA	NA
TOTAL AREA AVAILABLE	Atot	452.969	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (615.6 \cdot 72.5717 \cdot 1.0 + 2 \cdot 172.0 \cdot 72.5717 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 446.751 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 615.6(1.0 \cdot 74.0 - 1.0 \cdot 72.572) - 2 \cdot 172.0$$

$$(1.0 \cdot 74.0 - 1.0 \cdot 72.5717) \cdot (1 - 1.0)$$

$$= 8.793 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np})(t_n - t_{rn})fr_2$$

$$= (2 \cdot 185.0)(56.54 - 46.49)1.0$$

$$= 37.192 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_n/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(185.0, 200.0, 150.0)) \cdot (175.0 - 59.5) \cdot 1.0$$

$$= 405.984 \text{ cm}^2 \text{ Includes Bevel Area of } 59.604 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 49.4882 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 75.5717 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 75.5717 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(75.5717, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(49.4882, 11.3312)$$

$$= 49.4882 \text{ mm.}$$

Available Nozzle Neck Thickness = 59.5400 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 68.4,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 276.6 N./mm ²	Passed
Occasional	: 47.6,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 14.7,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govern. thk, tg = 77.0, tr = 72.572, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.981, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	25 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 25 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -59 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 191.09/255.10 = 0.749

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 97.36/0.5773$
 $= 168.6326 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 923.4000 mm.
 Parallel to Vessel Wall, opening length d 461.7000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 185.0000 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	297.849	297.849
Area Available in Shell	A1	4.396	4.396
Area Available in Nozzle Wall	A2	37.192	37.192
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	-0.000	-0.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	342.341	342.341
TOTAL AREA AVAILABLE	Atot	383.929	383.929

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: SH. OUT NOZ. S2

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (446.7513 - 8.7928 + 2 \cdot 172.0 \cdot 1.0 \cdot (1.0 \cdot 74.0 - 72.5717)) \cdot 138)$
 $= 623190.31 \text{ Kg}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (379.5324 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 535468.81 \text{ Kg}$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (379.5324 + 0.0 + 1. + (254.56)) \cdot 138$$

$$= 893674.62 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (379.5324 + 0.0 + 1. + 0.0 + (254.56)) \cdot 138$$

$$= 893674.62 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 959.5999 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 103932. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 393.8) \cdot (175.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 2096016. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 959.5999 \cdot (77.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 1161492. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (103932 + 2096016) = 2199948 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (103932 + 0 + 1161492 + 0) = 1265424 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (103932 + 1161492 + 0) = 1265424 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 2199948 Kgf, must exceed W = 623190 Kgf or W1 = 535468 Kgf
 Path 2-2 = 1265423 Kgf, must exceed W = 623190 Kgf or W2 = 893674 Kgf
 Path 3-3 = 1265423 Kgf, must exceed W = 623190 Kgf or W3 = 893674 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 192 bars

The Drop for this Nozzle is : 477.5000 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 754.5000 mm.

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: SH. OUT NOZ. S:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	955.000	mm.
Shell Thickness	77.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	609.600	mm.
Nozzle Thickness	175.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	191.09	bars
Radial Load	P	-19579.92	Kgf
Circumferential Shear	Vc	14684.89	Kgf
Longitudinal Shear	Vl	19579.92	Kgf
Circumferential Moment	Mc	30094.50	Kg-m.
Longitudinal Moment	Ml	30094.50	Kg-m.
Torsional Moment	Mt	35230.17	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (14.403) is < 20
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Dv (.865) is > 0.5
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (5.483) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 959.6 + 2 \cdot 1.65 \cdot \sqrt{(517.5(77.0 - 3.0))}$
 $= 1605.380 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1035.000 mm.
Nozzle Outside Diameter	(d) =	959.600 mm.
Vessel Thickness used	(T) =	74.000 mm.
Nozzle Thickness used	(t) =	172.000 mm.
T / t	=	0.430
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.467$$

Nr/P	=	0.059
Mr/P	=	0.090
M0/P	=	0.031
N0/P	=	0.079
MrD/Mc	=	0.260
NrDL/Mc	=	0.098
M0d/Mc	=	0.078
N0DL/Mc	=	0.086
MrD/Ml	=	0.086
NrDL/Ml	=	0.048
M0D/Ml	=	0.024
N0DL/Ml	=	0.086

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom



longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	115	134	115	134
Outplane Stress Summary	111	133	137	127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	57	57	57	57
Inplane Stress Summary	47	66	110	14

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	111	134	137	127

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-26	26	26	-26
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	57	57	57	57
Outplane Stress Summary	35	75	98	32

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-87	87	87	-87
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	115	134	115	134

Inplane Stress Summary	43	199	229	35
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	1	1	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	43	199	229	38
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	32	55	32	55

Hoop Stress Summary	30	52	39	62
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	16	16	16	16

Axial Stress Summary	10	13	26	14
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	30	53	39	62
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):



CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	32	55	32	55
Hoop Stress Summary	30	52	39	62

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-13	13	13	-13
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	1	23	36	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	2	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	30	52	40	62

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	115	134	115	134	115	134	115	134
Circ. Pl (SUS)	-2	-2	6	6	-3	-3	7	7
Circ. Q (SUS)	-2	2	14	-14	-68	68	106	-106
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	0	0	4	4	-2	-2	6	6
Long. Q (SUS)	-9	9	47	-47	-20	20	32	-32
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	115.0	134.0	115.0	134.0	115.0	134.0	115.0	134.0
Pm+Pl (SUS)	113.0	132.0	121.0	140.0	112.0	131.0	122.0	141.0
Pm+Pl+Q (Total)	111.3	134.2	135.1	126.0	44.4	199.0	228.1	37.5



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	134.00	138.00	Passed
Pm+Pl (SUS)	141.01	207.00	Passed
Pm+Pl+Q (TOTAL)	228.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	55	32	55	32	55	32	55
Circ. Pl (SUS)	-2	-2	6	6	-2	-2	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-4	-4	4	4	-4	-4	4	4
Long. Q (SUS)	-1	1	5	-5	-11	11	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	32.0	55.0	32.0	55.0	32.0	55.0	32.0	55.0
Pm+Pl (SUS)	30.0	53.0	38.0	61.0	30.0	53.0	38.0	61.0
Pm+Pl+Q (Total)	30.1	53.0	38.1	61.0	30.0	53.0	38.3	61.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	55.00	138.00	Passed
Pm+Pl (SUS)	61.00	207.00	Passed
Pm+Pl+Q (TOTAL)	61.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: TS. IN NOZ. T1

From: 50

Pressure for Reinforcement Calculations	P	147.094	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	955.00	mm.
Shell Finished (Minimum) Thickness	t	60.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		9758.17	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	40.4900	mm.
Flange Material		SA-350 LF2 +IT	
Hub Height of Integral Nozzle	h	145.0000	mm.
Height of Beveled Transition	L'	37.2448	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		105.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		60.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: TS. IN NOZ. T1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	16.000	in.
Actual Thickness Used in Calculation	1.594	in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 147.09 \cdot 480.5 / (138 \cdot 1.0 - 0.6 \cdot 147.09) \\
 &= 54.7200 \text{ mm.}
 \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 147.09 \cdot 206.2 / (138 \cdot 1.0 - 0.6 \cdot 147.09) \\
 &= 23.4823 \text{ mm.}
 \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$\begin{aligned}
 &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\
 &= 64.51 / 0.5773 \\
 &= 111.7346 \text{ mm.}
 \end{aligned}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	824.8000	mm.
Parallel to Vessel Wall, opening length	d	412.4000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	142.5000	mm.

Weld Strength Reduction Factor [fr1]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 138.0 / 138.0) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr2]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 138.0 / 138.0) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr3]:

$$\begin{aligned}
 &= \min(fr2, fr4) \\
 &= \min(1.0, 1.0) \\
 &= 1.000
 \end{aligned}$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	225.665	NA	NA
Area in Shell	A1	9.403	NA	NA
Area in Nozzle Wall	A2	39.922	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	183.854	NA	NA
TOTAL AREA AVAILABLE	Atot	234.178	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (412.4 \cdot 54.72 \cdot 1.0 + 2 \cdot 102.0 \cdot 54.72 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 225.665 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$



$$= 412.4(1.0 \cdot 57.0 - 1.0 \cdot 54.72) - 2 \cdot 102.0$$

$$(1.0 \cdot 57.0 - 1.0 \cdot 54.72) \cdot (1 - 1.0)$$

$$= 9.403 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np})(t_n - t_{rn})fr_2$$

$$= (2 \cdot 142.5)(37.49 - 23.48)1.0$$

$$= 39.922 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_n/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(142.5, 300.0, 145.0)) \cdot (105.0 - 40.5) \cdot 1.0$$

$$= 183.854 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 26.4823 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 57.7200 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 57.7200 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(57.72, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(26.4823, 11.3312)$$

$$= 26.4823 \text{ mm.}$$

Available Nozzle Neck Thickness = 40.4900 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 68.6,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 276.4 N./mm ²	Passed
Occasional	: 37.1,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 21.9,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govern. thk, tg = 60.0, tr = 54.72, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.96, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	19 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 19 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -70 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 147.09/255.10 = 0.577

Weld Size Calculations, Description: TS. IN NOZ. T1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (225.6654 - 9.4027 + 2 \cdot 102.0 \cdot 1.0 \cdot (1.0 \cdot 57.0 - 54.72)) \cdot 138)$
 $= 310860.53 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (223.7753 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 316294.16 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (223.7753 + 0.0 + 1. + (116.28)) \cdot 138$
 $= 479918.38 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (223.7753 + 0.0 + 1. + 0.0 + (116.28)) \cdot 138$
 $= 479918.38 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 616.4 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 66761. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 257.2) \cdot (105.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 811823. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 616.4 \cdot (60.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 574687. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (66761 + 811823) = 878584 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (66761 + 0 + 574687 + 0) = 641448 \text{ Kgf}$



Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (66761 + 574687 + 0) = 641448 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 878584 Kgf, must exceed W = 310860 Kgf or W1 = 316294 Kgf
 Path 2-2 = 641448 Kgf, must exceed W = 310860 Kgf or W2 = 479918 Kgf
 Path 3-3 = 641448 Kgf, must exceed W = 310860 Kgf or W3 = 479918 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 149 bars

The Drop for this Nozzle is : 112.7823 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 472.7823 mm.

Input Echo, WRC297 Item 1, Description: TS. IN NOZ. T1:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	955.000	mm.
Shell Thickness	60.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	406.400	mm.
Nozzle Thickness	105.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	147.09	bars
Radial Load	P	-13053.20	Kgf
Circumferential Shear	Vc	9789.91	Kgf
Longitudinal Shear	Vl	13053.20	Kgf
Circumferential Moment	Mc	13570.10	Kg-m.
Longitudinal Moment	Ml	13570.10	Kg-m.
Torsional Moment	Mt	15657.81	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dv/Tv (17.917) is < 20
 check the limitations of WRC bulletin 297.

Warning - The ratio, Dn/Dv (.573) is > 0.5
 check the limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (5.870) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

= NozzleOD + 2 • 1.65 • √(Rmean(t - ca))
 = 616.4 + 2 • 1.65 • √(509.0(60.0 - 3.0))

= 1178.496 mm.

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1018.000 mm.
 Nozzle Outside Diameter (d) = 616.400 mm.
 Vessel Thickness used (T) = 57.000 mm.
 Nozzle Thickness used (t) = 102.000 mm.
 T / t = 0.559
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.559$$

Nr/P = 0.066
 Mr/P = 0.118
 M0/P = 0.037
 N0/P = 0.091
 MrD/Mc = 0.269
 NrDL/Mc = 0.106
 M0d/Mc = 0.083
 N0DL/Mc = 0.077
 MrD/Ml = 0.102
 NrDL/Ml = 0.056
 M0D/Ml = 0.030
 N0DL/Ml = 0.109

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	117	131	117	131
Outplane Stress Summary	110	131	148	121

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-40	40	40	-40
Inplane Pressure Stress	58	58	58	58
Inplane Stress Summary	44	70	133	-3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top	Al Top	Bu Bottom	Bl Bottom
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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	110	132	149	125

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-33	33	33	-33
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58
Outplane Stress Summary	32	81	109	25

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-107	107	107	-107
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	117	131	117	131
Inplane Stress Summary	33	206	261	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	34	206	262	27

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	46	28	46
Hoop Stress Summary	24	42	39	57

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
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pipe cross-section)	Outside	Inside	Outside	Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-9	9	9	-9
Axial Pressure Stress	14	14	14	14

Axial Stress Summary	4	10	37	6
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	3	3	1	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	25	42	40	57
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	46	28	46

Hoop Stress Summary	27	44	37	54
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-26	26	26	-26
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	14	14	14	14

Axial Stress Summary	-12	28	55	-10
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	2	2	2	2

Shear Stress Summary	1	1	3	3
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stress Intensities Outside Inside Outside Inside

Two * Max Shear Stress 39 44 56 66

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	117	131	117	131	117	131	117	131
Circ. Pl (SUS)	-4	-4	10	10	-5	-5	9	9
Circ. Q (SUS)	-4	4	20	-20	-80	80	134	-134
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	-1	-1	5	5	-2	-2	8	8
Long. Q (SUS)	-13	13	67	-67	-25	25	41	-41
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	117.0	131.0	117.0	131.0	117.0	131.0	117.0	131.0
Pm+Pl (SUS)	113.0	127.0	127.0	141.0	112.1	126.1	126.1	140.1
Pm+Pl+Q (Total)	109.4	131.4	147.5	125.1	33.6	206.0	260.2	26.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+Pl (SUS)	141.01	207.00	Passed
Pm+Pl+Q (TOTAL)	260.23	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	28	46	28	46	28	46	28	46
Circ. Pl (SUS)	-4	-4	10	10	-2	-2	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-3	3	15	-15	-20	20	32	-32
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	28.0	46.0	28.0	46.0	28.0	46.0	28.0	46.0
Pm+Pl (SUS)	24.0	42.0	38.0	56.0	26.1	44.0	36.1	54.0
Pm+Pl+Q (Total)	24.2	42.1	39.2	56.1	39.1	44.1	53.5	65.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	138.00	Passed
Pm+Pl (SUS)	56.00	207.00	Passed
Pm+Pl+Q (TOTAL)	65.28	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: VENT NOZ. V1

From: 50

Pressure for Reinforcement Calculations	P	147.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	955.00	mm.
Shell Finished (Minimum) Thickness	t	60.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		9658.17	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	8.7376	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: VENT NOZ. V1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687	in.
Actual Thickness Used in Calculation	0.344	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147.0 \cdot 480.5 / (138 \cdot 1.0 - 0.6 \cdot 147.0)$
 $= 54.6828$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147.0 \cdot 24.42 / (138 \cdot 1.0 - 0.6 \cdot 147.0)$

= 2.7797 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	174.3250	mm.
Parallel to Vessel Wall	Rn+tn+t	87.1625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	14.3440	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: VENT NOZ. V1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.7797	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 57.6828	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 57.6828	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 6.4200	mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[6.42, max(57.6828, 4.5)]
 = 6.4200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(5.7797, 6.42)
 = 6.4200 mm.

Available Nozzle Neck Thickness = 0.875 * 8.738 = 7.645 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 7.645, tr = 2.78, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.598, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Weld Size Calculations, Description: VENT NOZ. V1

Intermediate Calc. for nozzle/shell Welds	Tmin	5.7376	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.0163 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 0.9536 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 360.9536 mm.

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Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: MAIN SHELL

Shell Maximum Design Pressure	Psd,max	191.00 bars
Shell Maximum Operating Pressure	Psox,max	191.00 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	77.0000 mm.
Shell Internal Corrosion Allowance	cas	3.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	955.000 mm.
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	200.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH. SHELL-1

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	147.00 bars
Channel Maximum Operating Pressure	Ptox,max	147.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	60.0000 mm.
Channel Internal Corrosion Allowance	cac	3.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	955.000 mm.
Channel Design Temperature	TEMPC	85.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	367
Tube Wall Thickness	et	3.4100 mm.
Tube Outside Diameter	D	25.4000 mm.
Total Straight Tube Length	Lt	7200.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		6840.00 mm.
Design Temperature of the Tubes		200.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	154.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	39.3700 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm.
Groove Weld Leg	ag	3.0000 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	438.406 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	4389.4 cm ²
Perimeter of Tube Layout	Cp	3589.6 mm.

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 200.00 °C
 Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 180.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 3.0000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm.
 Tubesheet Outside Diameter A 1109.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 1608.000 mm.
 Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 88.22 °C
 Mean Tube Metal Temp. along Tube length Ttm 66.48 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	88.2 °C	0.0000120258 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.19258E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	88.2 °C	0.19886E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	66.5 °C	0.0000118850 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.19258E+09 KPa.
Elastic Mod. at Metal Temp. along Len	66.5 °C	0.20010E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	200.0 °C	0.19258E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 200.0 °C 0.19258E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (147.0 \cdot (25.4/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 147.0)$
 $= 1.8997 + 0.0000 = 1.8997 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 200.00 °C
 Elastic Modulus for Material 193602528.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
3.4100	25.40	6840.00	7.45	50.0000	0.0198260	93.30

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 193.7103 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
3.3762	25.40	6840.00	7.52	50.0000	0.0194353	93.30

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 191.0229 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	3.3762 mm.
Allowable Internal Pressure at Corroded thickness	277.93 bars
Required Internal Design Pressure	147.00 bars
Allowable External Pressure at Corroded thickness	193.71 bars
Required External Design Pressure	191.00 bars
Required Thickness due to Shell Side pressure	3.3762 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$= l_{tx} / h$ (modified for corrosion if present)
 $= 0/180.0$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d*]:

$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$
 $= \max(25.4 - 2 \cdot 3.41 (0.19258E+09/0.19258E+09) \cdot$
 $(92/137) \cdot (0); 25.4 - 2 \cdot 3.41)$
 $= 25.4000 \text{ mm.}$

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot ro + dt$
 $= 2 \cdot 438.406 + 25.4$
 $= 902.212 \text{ mm.}$

Determine the basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (39.37 - 25.4) / 39.37$
 $= 0.355$



Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 902.212 / 2$$

$$= 451.1060 \text{ mm.}$$

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2))}$$

$$= 39.37 / \sqrt{(1 - 4 \cdot \min(0 \cdot 100.0; 4 \cdot 902.212 \cdot 39.37) / (\pi \cdot 902.212^2))}$$

$$= 39.3700 \text{ mm.}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (39.37 - 25.4) / 39.37$$

$$= 0.3548$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 477.5 / 451.106$$

$$= 1.0585$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 477.5 / 451.106$$

$$= 1.0585$$

Compute parameter [xt]:

$$= 1 - Nt \cdot ((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 367 \cdot ((25.4 - 2 \cdot 3.41) / (2 \cdot 451.106))^2$$

$$= 0.8444$$

Determine parameter [xs]:

$$= 1 - Nt \cdot (dt / (2 \cdot a_o))^2$$

$$= 1 - 367 \cdot (25.4 / (2 \cdot 451.106))^2$$

$$= 0.7091$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s (D_s + t_s) E_s / L$$

$$= \pi \cdot 77.0 (955.0 + 77.0) \cdot 19258E+09 / 6840.0$$

$$= 716760.0000 \text{ Kg/mm.}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 3.41 (25.4 - 3.41) \cdot 19258E+09 / 6840.0$$

$$= 676.3679 \text{ Kg/mm.}$$

Stiffness Factor [Ks,t]:

$$= K_s / (Nt \cdot K_t)$$

$$= 716760 / (367 \cdot 676.368)$$

$$= 2.88752$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 716760 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((955.0 + 77.0) 77.0)$$

$$= 0.0064 \text{ 1/mm.}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.006 \cdot 0.19258E+09 \cdot 77.0^3 / (6(1 - 0.3^2))$$

$$= 1038395968.0000 \text{ bars} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(955.0 + 77.0) \cdot 0.10384E+1 / 180.0^3 \cdot (1 + 180.0 \cdot 0.006 + (180.0 \cdot 0.006)^2 / 2)$$

$$= 3124959.7500 \text{ bars}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 955.0^2 / (4 \cdot 0.19258E+09 \cdot 77.0) \cdot (1 - 955.0 \cdot 0.3 / (2(955.0 + 77.0)))$$

$$= 0.13241E-01 \text{ mm./N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / ((D_c + t_c) t_c)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((955.0 + 60.0) 60.0)$$

$$= 0.0074 \text{ 1/mm.}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_c^2))$$

$$= 0.007 \cdot 0.19904E+09 \cdot 60.0^3 / (6(1 - 0.3^2))$$

$$= 580033856.0000 \text{ bars} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(955.0 + 60.0 / \cos(0)) \cdot 0.58003E+09 / 180.0^3 \cdot (1.0 + 180.0 \cdot 0.007 \cdot (1.0 + 1.0) / 2 + (180.0 \cdot 0.007)^2 / 2 \cdot 1.0)$$

$$= 1941229.0 \text{ bars}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 955.0^2 / (4 \cdot 0.19904E+09 \cdot 60.0) \cdot (1 - 955.0 \cdot 0.3 / (2(955.0 + 60.0)))$$

$$= 0.16396E-01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 4.572009 ; \mu^* = 0.354839 ; E^* = 68094424 \text{ KPa.}$$

$$E^*/E = 0.353589 ; \nu^* = 0.324557$$

Note: As h/p (4.572) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^{*2}) Nt \cdot Et \cdot t(dt - tt) a o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.325^2) 367 \cdot 0.19258E+09 \cdot 3.41 (25.4 - 3.41) 451.106^2) / (68094424 \cdot 6840.0 \cdot 180.0^3))^{1/4}$$

$$= 1.7087$$

Values from Table 4.18.3

$$Z_d = 0.287102 ; Z_v = 0.173474 ; Z_m = 0.709056$$

$$Z_a = 0.629339E+00 ; Z_w = 0.173474$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1109.0/902.212$$

$$= 1.2292$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32) / 68094424 \cdot (3124959 + 1941229 + 0.19258E+09 \cdot \ln(1.23))$$

$$= 5.4195$$

Compute parameter [φ]:

$$= (1 + v \cdot F)$$

$$= (1 + 0.3246) \cdot 5.4195$$

$$= 7.1784$$

Compute parameter [Q1]:

$$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.0585 - 1 - 7.1784 \cdot 0.1735) / (1 + 7.1784 \cdot 0.7091)$$

$$= -0.194874$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.2871 + -0.19487 \cdot 0.17347) / 2 \cdot 1.70874^4$$

$$= 1.0797$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.17347 + -0.19487 \cdot 0.70906) / 2 \cdot 1.70874^4$$

$$= 0.1505$$

Compute parameter [U]:

$$= (Z_w + (p_s - 1) \cdot Z_m) \cdot X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.1735 + (1.0585 - 1) \cdot 0.7091) \cdot 1.70874^4 / (1 + 7.1784 \cdot 0.7091)$$

$$= 0.3009$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm. (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.0585 \cdot 0.10384E+1 \cdot 0.0064 \cdot 0.013241 (1 + 180.0 \cdot 0.0064)$$

$$= 20280.4688 \text{ mm.}^2$$

Calculate parameter [ω*s]:

$$= a_o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s$$

$$= 451.106^2 (1.059^2 - 1) \cdot (1.059 - 1) / 4 - 20280.469$$

$$= -19921.9570 \text{ mm.}^2$$

Parameter [wc]:

$$= p_c \cdot k_c [\delta_c (C_m \theta + \beta_c + C_q \Delta \cdot h \cdot \beta_c^2) - \zeta_c (C_m \theta + h \cdot C_q \theta + h \cdot \beta_c / 2)]$$

$$= 1.059 \cdot 0.58003E+09 [0.016396 (1.0 + 0.007 \cdot 1.0 \cdot 180.0) - 0 (1.0/180.0 + 1.0 \cdot 0.007/2)]$$

$$= 17249.1328 \text{ mm.}^2$$

Parameter [ω*c]:

$$= a_o^2 [(p_c^2 + 1) \cdot (p_c - 1) / 4 - (p_s - 1) / 2] - \omega_c$$

$$= 451.10599^2 [(1.05851^2 + 1) \cdot (1.05851 - 1) / 4 - (1.05851 - 1) / 2] - 17249.13281$$

$$= -16890.6211 \text{ mm.}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/K_{st} \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1) / (J \cdot K_{st})] - [(1 - J) / (2J \cdot K_{st})] [(D_j^2 - (D_s)^2) / D_o^2] \}$$

$$= 191.0 \{ 0.709 + 2(1 - 0.709) \cdot 0.3 + [2/2.888 \cdot (955.0/902.212)^2] \cdot 0.3 - [(1.059^2 - 1) / (1.0 \cdot 2.888)] - [(1 - 1.0) / (2 \cdot 1.0 \cdot 2.888)] [(0^2 - (955.0)^2) / 902.212^2] \}$$



= 205.2780 bars

Pressure [P't]:

$$= [xt + 2(1 - xt)vt + 1/(J \cdot Kst)]Pt$$

$$= [0.844 + 2(1 - 0.844)0.3 + 1/(1 \cdot 2.888)]147.0$$

$$= 188.7567 \text{ bars}$$

Pressure [P_γ]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 367 \cdot 676.368 \cdot 0 / (3.142 \cdot 451.106^2)$$

$$= 0.0000 \text{ bars}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W^* / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 0.301 \cdot 0 / (2 \cdot \pi \cdot 451.106^2)$$

$$= -0.0000 \text{ bars}$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega^*s \cdot Ps - \omega^*c \cdot Pt)$$

$$= -(0.301/451.106^2) \cdot (-19921.957 \cdot 191.0 - -16890.621 \cdot 147.0)$$

$$= 1.9552 \text{ bars}$$

Calculate the pressure [P_w]:

$$= (U/ao^2) \cdot (\omega^*s \cdot P^*s - \omega^*c \cdot P^*c)$$

$$= (0.301/451.106^2) \cdot (20280.4688 \cdot 0 - 17249.1328 \cdot 0)$$

$$= 0.0000 \text{ bars}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (P^*s - P^*t + P_\gamma + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 2.888 / (1 + 1.0 \cdot 2.888(1.08 + (1.059 - 1)0.15)) \cdot (205.278 - 188.757 + 0 + 0 + 0 + 1.955)$$

$$= 12.8772 \text{ bars}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega^*s \cdot Ps - \omega^*c \cdot Pt) - (\omega^*s \cdot P^*s - \omega^*c \cdot P^*c)) + W^* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-19921.957 \cdot 191.0 - -16890.621 \cdot 147.0) - (20280.469 \cdot 0 - 17249.133 \cdot 0)) \cdot 0.01 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 7.17839 \cdot 0.70906)$$

$$= -2213.9902 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.195 + 2 \cdot -2213.99 / (12.877 \cdot 451.106^2)$$

$$= -0.36058$$

Fm Value from Table 4.18.3 = 0.180288

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu^*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.1803 / 0.3548) \cdot (2 \cdot 451.106 / (180.0 - 0))^2 \cdot 12.88$$

$$= 24.6573 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138.0 = 207.0 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σf]:

$$= (1.5 \cdot Fm / \mu^*) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.0378 / 0.3548) \cdot (2 \cdot 451.106 / (16.671 - 0))^2 \cdot 4.42$$

$$= 206.9949 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 16.6705 + 0 + 0$$

$$= 16.6705 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [Tlimit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.355 \cdot 180.0 / 451.106) \cdot 110.398$$

$$= 31.2620 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm.}$$

$$\text{Tubesheet Shear Stress} = 14.6234 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(16.6705; 3.81)$$

$$= 16.6705 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = 0.9572, F_{tmax} = 1.0143$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((191.0 \cdot 0.7091 - 147.0 \cdot 0.8444) - (12.877) \cdot 0.957) /$$

$$(0.8444 - 0.7091)$$

$$= -0.7424 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((191.0 \cdot 0.7091 - 147.0 \cdot 0.8444) - (12.877) \cdot 1.014) /$$

$$(0.8444 - 0.7091)$$

$$= -1.2864 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 1.286 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 92.3983 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.287 + -0.361 \cdot 0.173)1.709^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2)$$

Determine the factor [rt]:

$$= (\sqrt{dt^2 + (dt - 2 \cdot tt)^2}) / 4$$

$$= (\sqrt{25.4^2 + (25.4 - 2 \cdot 3.41)^2}) / 4$$

$$= 7.8676 \text{ mm.}$$

Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{yt})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19258E+09 / 153)}$$

$$= 157.1179$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1.0 \cdot 1608.0 / 7.868$$

$$= 204.3835$$

Buckling Allowable Stress [Stb]:

$$= (1 / F_s) \cdot (\pi^2 \cdot E_t / F_t^2) (C_t \leq F_t)$$

$$= (1 / 2.0) \cdot (\pi^2 \cdot 0.19258E+09 / 204.3835^2)$$

$$= 22.752 \text{ N./mm}^2 \text{ (Never greater than } S_{ot})$$

Note: The Axial Compressive stress in tubes is within limits.



Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 1.29 \cdot 2.3558$$

$$= 30.90 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 3.41) 3.41$$

$$= 2.356 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.41 (25.4 - 3.41) 92.398$$

$$= 2219.550 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3.0 (25.4 + 0.67 \cdot 3.0) 92.4$$

$$= 2068.8767 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92.398 / 92.398$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3.0 (25.4 + 0.67 \cdot 3.0) 92.4$$

$$= 1338.6848 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 3.41 \cdot (25.4 - 3.41) 1.0 \cdot 1.0)} - 0.75 \cdot 25.4 \right)$$

$$= 4.002 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 2.0460 mm.

Minimum Required Groove Weld Leg [agr]: 2.0460 mm.

Tube-Tubesheet Jt allowable, 2219.55 is $\geq$ tube strength 2219.55 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 138.0 \cdot 1.0 = 138.0 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s ,m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 451.106^2 / ((955.0 + 77.0) 77.0) \cdot [12.88 + (1.059^2 - 1)(191.0 - 147.0)] + 477.5^2 \cdot 147.0 / ((955.0 + 77.0) 77.0)$$

$$= 46.8363 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 77.0 mm., the minimum Shell length adjacent to the tubesheet should be 488.112 mm.

Shell Membrane Stress due to Joint Interaction [σ_s ,m]:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 451.106^2 / ((955.0 + 77.0) 77.0) [12.88 + (1.059^2 - 1)(191.0 - 147.0)] + 477.5^2 \cdot 147.0 / ((955.0 + 77.0) 77.0)$$

= 46.8363 N./mm²

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E^*) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 0.10384 E + 1 / 77.0^2 \{ 0.006 [0.013241 \cdot 191.0 + 477.5^2 \cdot 0 / (.14829 E + 11)] + 6(1 - 0.32^2) / (68094424) (451.11 / 180.0)^3 (1 + 180.0 \cdot 0.01 / 2) [12.9 (0.173 + 0.709 \cdot -0.195) + 2 / 451.11^2 \cdot 0.709 \cdot -2213.99] \}$$

= 149.5853 N./mm²

Shell Stress Summation vs. Allowable

$$\therefore | \sigma_{s,m} | + | \sigma_{s,b} | \leq 1.5 \cdot S_s$$

$$\therefore | 46.84 | + | 149.59 | \leq 207.0 \text{ N./mm}^2$$

$$\therefore 196.42, \text{ must be } \leq 207.0 \text{ N./mm}^2$$

Note:

For a given Channel thickness of 60.0 mm., the minimum Channel length adjacent to the tubesheet should be 430.874 mm.

Channel Membrane Stress due to Joint Interaction [σ_c, m]:

$$= a c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$$

$$= 477.5^2 \cdot 147.0 / ((955.0 + 60.0) 60.0 \cdot \cos(0))$$

= 55.039 N./mm²

Channel Bending Stress due to Joint Interaction [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^2) / E^*) \cdot (a_o / h)^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 0.58003 E + 09 / 60.0^2 \cdot \{ (0.016396 \cdot 1.0 \cdot 0.007 - 0 \cdot 1.0) \cdot 147.0 - 6((1 - 0.325^2) / 68094424) \cdot (451.106 / 180.0)^3 \cdot (1.0 + 180.0 \cdot 0.007 / 2 \cdot 1.0) \cdot [12.9 (0.17347 + 0.70906 \cdot -0.195) + 2 / 451.106^2 \cdot 0.70906 \cdot -2213.99] \}$$

= 192.7746 N./mm²

Channel Stress Summation vs. Allowable

$$\therefore | \sigma_{c,m} | + | \sigma_{c,b} | \leq 1.5 \cdot S_c$$

$$\therefore | 55.04 | + | 192.77 | \leq 207.0 \text{ N./mm}^2$$

$$\therefore 247.81 \text{ must be } \leq 207.0 \text{ N./mm}^2 \text{ (See Elast-Plast iteration below)}$$

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	24.7 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	2.5 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	1.3 <=	92.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-1.3 <=	-22.8 N./mm ²	Pass
Maximum Force on any one Tube	30.9 <=	2219.6 Kg _f	Pass
Axial Membrane Stress in Shell	46.8 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	196.4 <=	207.0 N./mm ²	Pass
Channel Stress (jt. inter.)	247.8 <=	207.0 N./mm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [S_{stars}]:

$$= \min(S_{ys}, S_{PS}, s/2)$$

$$= \min(225, 450/2) = 225.01 \text{ N./mm}^2$$

Compute Allowable [S^*c]:

$$= \min(S_{yc}, S_{PS}, c/2)$$

$$= \min(242, 484/2) = 242.01 \text{ N./mm}^2$$

Compute Factor [facts]:
 $= \min([1.4 - 0.4 \cdot \text{abs}(\sigma, b) / S_{\text{stars}}], 1)$
 $= \min([1.4 - 0.4 \cdot \text{abs}(149.59) / 225.01], 1) = 1.0$

Compute Factor [factc]:
 $= \min([1.4 - 0.4 \cdot \text{abs}(\sigma, b) / S_{\text{starc}}], 1)$
 $= \min([1.4 - 0.4 \cdot \text{abs}(192.77) / 242.01], 1) = 1.0$

Note: Since facts & factc are both ≥ 1 , the design Passes.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Shell Stress (jt. inter.)	196.4	$\leq 207.0 \text{ N./mm}^2$	Pass
Channel Stress (jt. inter.)	247.8	$\leq 484.0 \text{ N./mm}^2$	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	16.671	180.000	Ok
Tube-Tubesheet Fillet Weld Leg :	2.046	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	2.046	3.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	156.963	...	160	207	26	110	Ps+Pt-Th	D1 Ok
D2uc	169.021	...	185	207	28	110	Ps+Pt-Th	D2 Ok
D3uc	16.671	...	25	207	2	110	Ps+Pt-Th	D3 Ok
O1uc	123.035	...	210	426	35	110	Ps+Pt+Th	O1 Ok
O2uc	84.241	...	134	426	19	110	Ps+Pt+Th	O2 Ok
O3uc	23.809	...	48	426	7	110	Ps+Pt+Th	O3 Ok
O4uc	37.191	...	52	426	9	110	Ps+Pt+Th	O4 Ok
D1c	161.667	...	168	207	26	110	Ps+Pt-Th-c	D1 Ok
D2c	173.558	...	193	207	29	110	Ps+Pt-Th-c	D2 Ok
D3c	21.673	...	25	207	2	110	Ps+Pt-Th-c	D3 Ok
O1c	127.854	...	220	426	36	110	Ps+Pt+Th-c	O1 Ok
O2c	89.904	...	141	426	19	110	Ps+Pt+Th-c	O2 Ok
O3c	29.879	...	51	426	7	110	Ps+Pt+Th-c	O3 Ok
O4c	42.631	...	54	426	9	110	Ps+Pt+Th-c	O4 Ok
Max:	173.5583	... mm.	0.934		0.325	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	3	137	...	...	:	...	...	...	...	:	...	Ok
D2uc	44	137	...	...	:	...	...	...	...	:	...	Ok
D3uc	47	137	...	...	:	...	...	...	...	:	...	Ok
O1uc	9	450	-9	-117	:	...	...	...	...	:	...	Ok
O2uc	31	450	...	...	:	...	...	...	...	:	...	Ok
O3uc	34	450	...	...	:	...	...	...	...	:	...	Ok
O4uc	12	450	-12	-117	:	...	...	...	...	:	...	Ok
D1c	4	137	...	...	:	...	...	...	...	:	...	Ok
D2c	46	137	...	...	:	...	...	...	...	:	...	Ok
D3c	49	137	...	...	:	...	...	...	...	:	...	Ok
O1c	9	450	-9	-117	:	...	...	...	...	:	...	Ok
O2c	33	450	...	...	:	...	...	...	...	:	...	Ok
O3c	36	450	...	...	:	...	...	...	...	:	...	Ok
O4c	13	450	-13	-117	:	...	...	...	...	:	...	Ok
Max RATIO												
	0.356		0.108			...			...			

Tube, Shell and Channel Stress Summary:

---- Tube Stresses						Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd		Stress	Allwd	Stress	Allwd	Fail
D1uc	13	92	-2	-22	315	2219		42	206	264	484	Ok
D2uc	14	92	-14	-22	333	2219		232	450	17	206	Ok
D3uc	1	92	-1	-22	31	2219		196	206	248	484	Ok
O1uc	52	184	...	...	1247	4439		84	450	299	484	Ok
O2uc	33	184	...	...	791	4439		184	450	18	484	Ok
O3uc	38	184	...	...	915	4439		149	450	282	484	Ok
O4uc	39	184	...	...	932	4439		48	450	34	484	Ok
D1c	13	92	-4	-22	319	2219		51	206	286	484	Ok
D2c	13	92	-13	-22	323	2219		249	450	23	206	Ok
D3c	1	92	-1	-22	19	2219		206	206	264	484	Ok
O1c	52	184	...	...	1244	4439		95	450	324	484	Ok
O2c	34	184	...	...	819	4439		198	450	15	484	Ok
O3c	38	184	...	...	921	4439		154	450	301	484	Ok
O4c	38	184	...	...	925	4439		51	450	37	484	Ok
Max RATIO												
	0.281		0.609		0.281			0.993		0.669		

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	173.558	180.000	Ok
Tube thickness :	3.376	3.410	Ok
Tube-tubesheet fillet weld leg :	2.046	3.000	Ok
Tube-tubesheet groove weld leg :	2.046	3.000	Ok

Min Shell length of thk, (77.000) adj. to tubesheet: 488.112 mm.

Min Channel length of thk, (60.000) adj. to tubesheet: 430.874 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.834 mm.
 Due to Pressure Shell Compresses by : -0.702 mm.
 Due to Pressure + Thermal Shell Compresses by : -2.535 mm.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	181.658	1.000	204.500	1.000
Tubesheet Shear Stress	561.754	1.000	731.500	1.000
Tube Tensile Stress	1021.58	1.000	1274.65	1.000
Tube Compressive Stress	803.686	1.000	313.865	1.000
Tube-Tubesheet Joint load	1021.58	1.000	1274.65	1.000
Shell Stress (Axial, Junction)	181.658	1.000	193.709	0.561
Tubesheet-Channel Junction	181.658	1.000	193.709	0.111
Tube Pressure Stress	277.928	1.000	193.709	1.000
Minimum MAWP	181.658		193.709	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	190.281	1.000	214.106	1.000
Tubesheet Shear Stress	632.090	1.000	749.283	1.000
Tube Tensile Stress	1036.52	1.000	1274.61	1.000
Tube Compressive Stress	1413.08	1.000	330.349	1.000
Tube-Tubesheet Joint load	1036.52	1.000	1274.61	1.000
Shell Stress (Axial, Junction)	190.281	1.000	198.975	0.465
Tubesheet-Channel Junction	190.281	1.000	198.975	0.082
Tube Pressure Stress	277.928	1.000	198.975	1.000
Minimum MAPnc	190.281		198.975	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm.)

Load Case#	Due to Pressure UnCorroded	Due to Pressure Corroded	Due to Thermal Exp. UnCorroded	Due to Thermal Exp. Corroded	Computed UnCorroded	Delta Ms Corroded
1	0.0959	0.0960	...	...	...	...
2	-0.7012	-0.7018	...	...	...	...
3	-0.6053	-0.6059	...	...	...	...
4	...	...	...	...	...	...
5	0.0959	0.0960	-1.8320	-1.8336	...	...
6	-0.7012	-0.7018	-1.8320	-1.8336	...	...
7	-0.6053	-0.6059	-1.8320	-1.8336	...	...
8	...	...	-1.8320	-1.8336	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(180.0/4, \min(180.0, 77.0)) \\
 &= 77.000 \text{ mm.}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(ps/\text{Shellside MAPnc}, 1) \\
 &= \min(191.0/198.98, 1) \\
 &= 0.960
 \end{aligned}$$

Thickness Ratio = 0.96, Temperature Reduction per Fig. UCS 66.1 = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	26 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	7 °C

Channel Side MDMT calculation:



Governing Channel side thickness per figure UCS-66.3 (e):
= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(180.0/4, min(180.0, 60.0))
= 60.000 mm.

MDMT reduction ratio per UCS 66 (b)(1)(b):
= min(pt/Tubeside MAPnc, 1)
= min(147.0/190.28, 1)
= 0.773

Thickness Ratio = 0.773, Temperature Reduction per Fig. UCS 66.1 = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	21 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-9 °C

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